CHAPTER

FOUR

Alternating Current Circuits

Lesson 1 Alternating Current Circuits.

Lesson 2 Following AC Circuits.

Lesson 3
- The Oscillating Circuit.
- The Resonant Circuit.

- Questions on each lesson from the Previous Years Exams.
- Test on Chapter 4

CamScanner

Scanned with OKEN Scanner

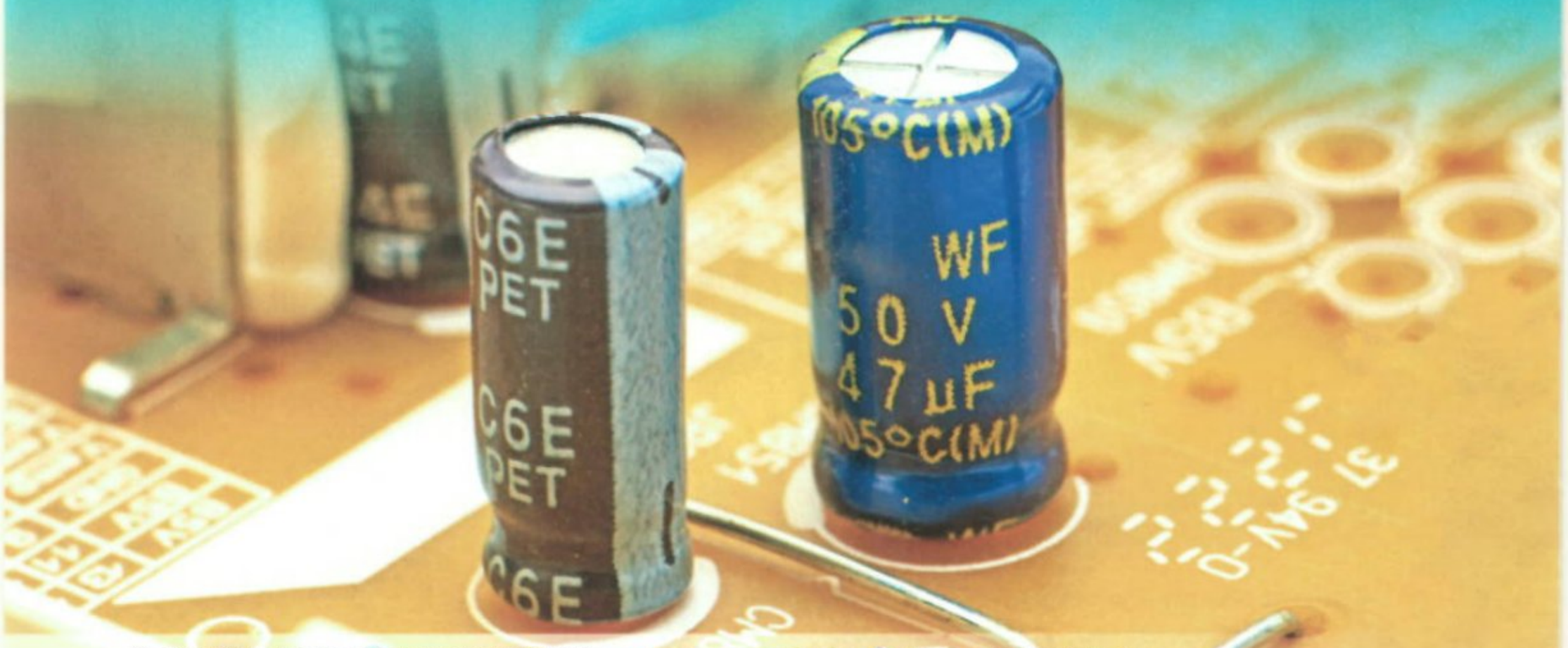
CamScanner

Chapter 4

Questions on Lesson One

Alternating Current Circuits

Interactive test



The questions signed by ✳ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

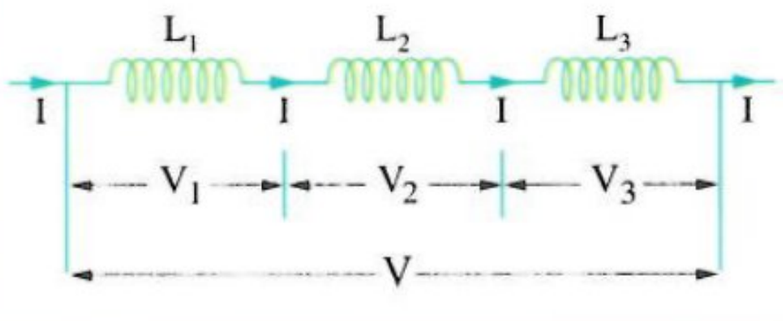
Important guidelines

Inductor

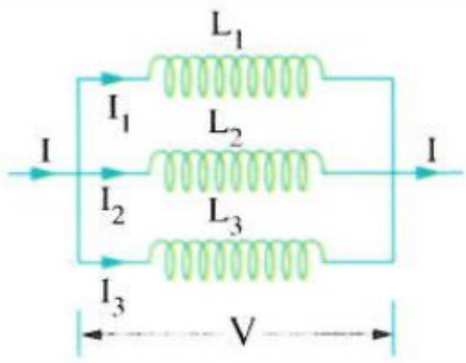
Self inductance of a solenoid	The value of AC current passing in a coil	Inductive reactance of a coil	Comparing inductive reactances of two coils
$L = \frac{\mu A N^2}{\ell}$	$I = \frac{V_L}{X_L}$	$X_L = \omega L = 2\pi f L$	$\frac{(X_L)_1}{(X_L)_2} = \frac{\omega_1 L_1}{\omega_2 L_2} = \frac{f_1 L_1}{f_2 L_2}$

Inductors connection

In series



In parallel



Scanned with OKEN Scanner

Then

$$L = L_1 + L_2 + L_3 + \cdots \qquad \frac{1}{L} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \cdots$$

$$X_L = (X_L)_1 + (X_L)_2 + (X_L)_3 + \cdots \qquad \frac{1}{X_L} = \frac{1}{(X_L)_1} + \frac{1}{(X_L)_2} + \frac{1}{(X_L)_3} + \cdots$$

If the coils are identical and their number is (n)

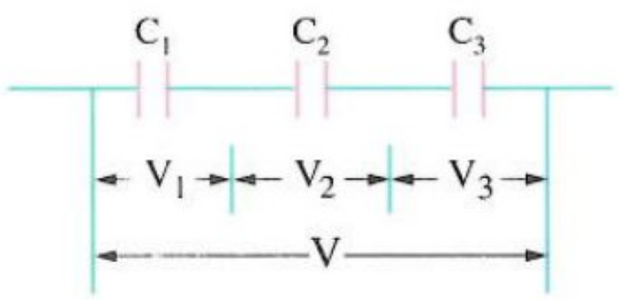
$$L = nL_1 \ , \ X_L = n\,(X_L)_1 \qquad L = \frac{L_1}{n} \ , \ X_L = \frac{(X_L)_1}{n}$$

Capacitor

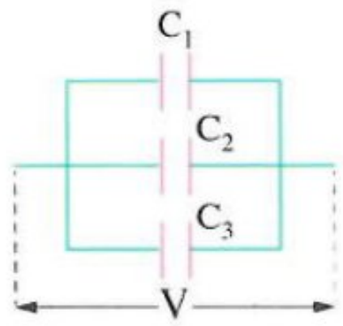
Capacitance of a capacitor	The value of AC current passing in a capacitor's circuit	Capacitive reactance of a capacitor	Comparing capacitive reactances of two capacitors
$C = \frac{Q}{V}$	$I = \frac{V_C}{X_C}$	$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$	$\frac{(X_C)_1}{(X_C)_2} = \frac{\omega_2 C_2}{\omega_1 C_1} = \frac{f_2 C_2}{f_1 C_1}$

Connection of capacitors

In series



In parallel



Then

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \qquad C = C_1 + C_2 + C_3 + \cdots$$

$$X_C = (X_C)_1 + (X_C)_2 + (X_C)_3 + \cdots \qquad \frac{1}{X_C} = \frac{1}{(X_C)_1} + \frac{1}{(X_C)_2} + \frac{1}{(X_C)_3} + \cdots$$

If the capacitors are identical and their number is (n)

$$C = \frac{C_1}{n} \qquad C = nC_1$$

$$X_C = n\,(X_C)_1 \qquad X_C = \frac{(X_C)_1}{n}$$

CamScanner

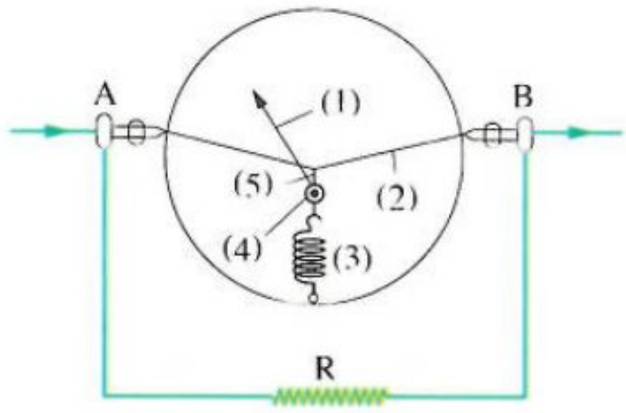
Scanned with OKEN Scanner

CamScanner

First Multiple choice questions

Hot-wire ammeter

1 The opposite figure represents the structure of one of the electric measuring devices, so.



(i) The idea of working of this device depends on .....
(a) the eddy currents
(b) the magnetic effect of electric current
(c) the thermal effect of electric current
(d) the electromagnetic induction

(ii) The resistance (R) is used for ..... .
(a) increasing the speed of heating the wire
(b) making the silk thread always stretched
(c) increasing the measuring range of the current intensity
(d) increasing the resistance of the device

(iii) The component that is made of platinum and iridium is ..... .
(a) component (1) (b) component (2) (c) component (3) (d) component (4)

(iv) The function of component (3) is ..... .
(a) making the device scale uniform
(b) increasing the measuring range of the device
(c) making the silk thread always stretched
(d) heating the device when an electric current passes through it

(v) The electric current passes through ..... .
(a) component (1) and component (3) (b) component (2) and component (5)
(c) component (1) and resistance R (d) component (2) and resistance R

2 In a hot wire ammeter, the thermal power generated in the platinum-iridium wire as the alternating electric current flows is directly proportional to the ..... .
(a) effective value of the current
(b) reciprocal of the effective value of the current
(c) maximum value of the current
(d) square of the effective value of the current

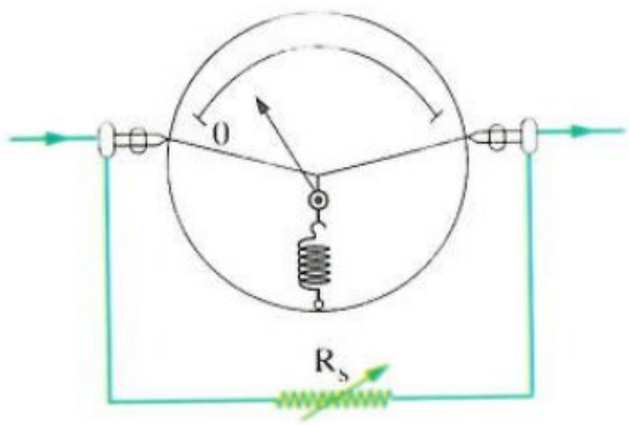
3 A thermal power is produced in a hot wire ammeter when AC current of maximum intensity 14 A passes through it. Then, the DC current intensity which produces the same thermal power in the same wire is approximately equal to ..... .
(a) 7 A (b) 10 A (c) 14 A (d) 20 A

Scanned with OKEN Scanner

4 A hot wire ammeter containing a shunt resistance in parallel with the platinum-iridium wire is connected in a circuit that carries AC current of effective value I, if the shunt resistance is increased while the same current (I) is passing in the circuit then the produced thermal power in the wire ..... .

(a) increases
(b) decreases
(c) does not change
(d) we can't determine the answer

5 In the opposite circuit, a hot-wire ammeter through which an electric current of intensity I passes. When the value of the shunt resistor (R_s) is decreased and the same current I get to pass through the ammeter, what happens to:



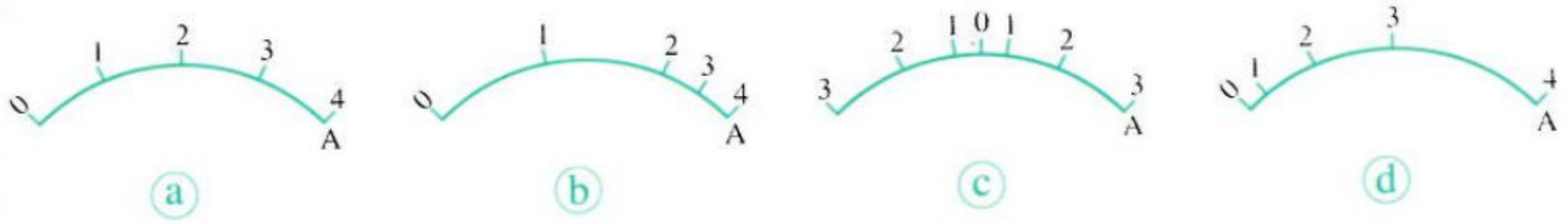
(i) the range and the accuracy of the ammeter reading, respectively?

(a) Decreases, decreases.
(b) Decreases, increases.
(c) Increases, decreases.
(d) Increases, increases.

(ii) the tension force in the spring coil?

(a) Increases.
(b) Decreases.
(c) Does not change.
(d) Cannot be determined.

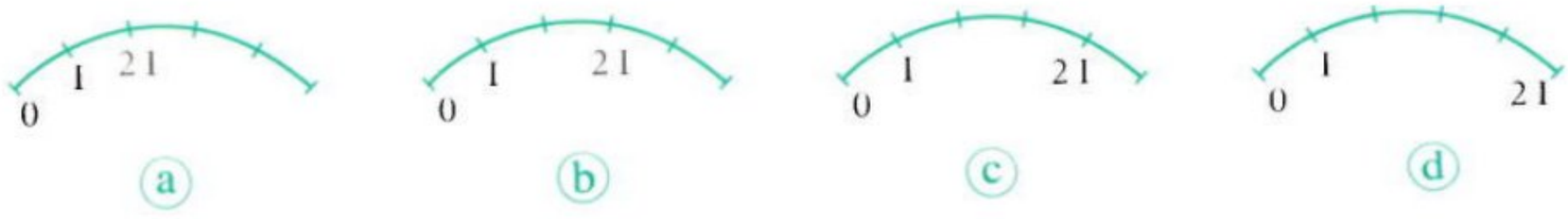
6 Which of the following figures represents correctly the scale of a hot wire ammeter?



(a) (b) (c) (d)

7 If when an AC current of effective value I passes through a hot wire ammeter device, its pointer gets deflected to the position shown in the opposite figure, so which one of the following figures represents the correct position of the pointer of the hot wire ammeter when the effective value of the passing current becomes 2 I?



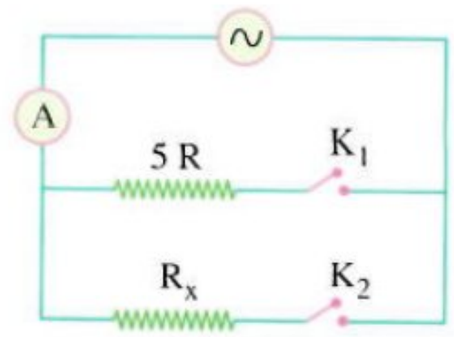


(a) (b) (c) (d)

CamScanner
Scanned with OKEN Scanner

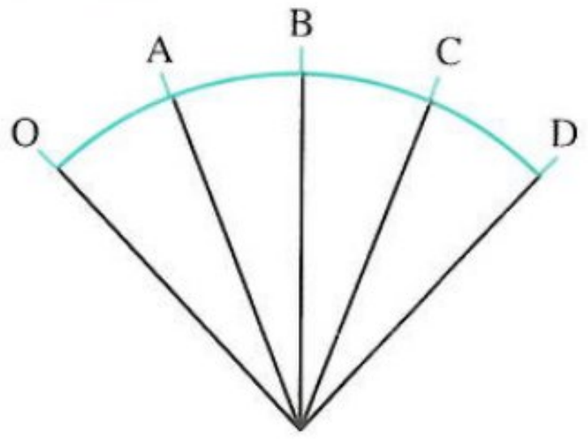
CamScanner

8 In the shown electric circuit, a hot-wire ammeter of resistance R is connected to an AC source of effective voltage V. When only switch K_1 is closed, the ammeter pointer deflects by an angle of θ from the zero position and when only switch K_2 is closed, the ammeter pointer deflects by an angle of 9 θ from the zero position. So, resistance R_x equals ..... .



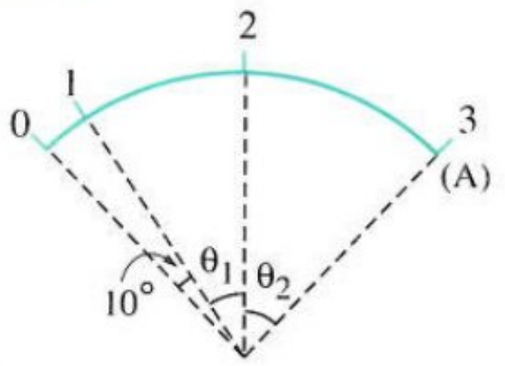
(a) 4 R (b) 2 R (c) R (d) 0.5 R

9 A student divided the space between the zero position (O) and the maximum possible reading (D) on a hot-wire ammeter scale into four equal divisions as shown in the figure. The division that indicates the largest change in the current value is ..... .



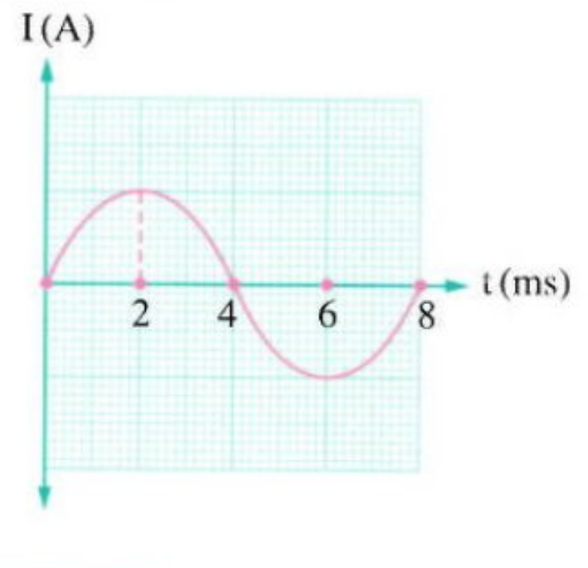
(a) OA (b) AB
(c) BC (d) CD

10 The opposite figure represents the scale of a hot-wire ammeter, so the values of the two angles θ_1 and θ_2 shown in the figure are respectively ....., ...



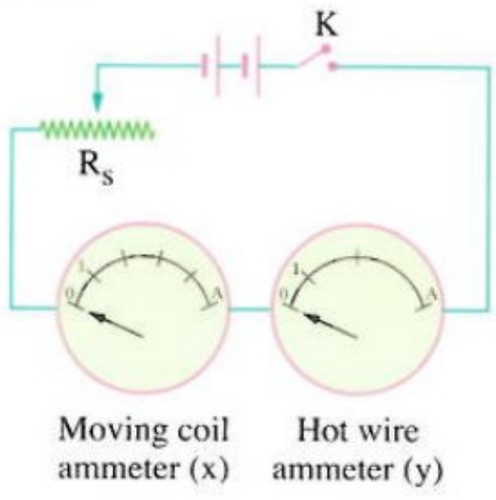
(a) 20°, 40° (b) 40°, 50°
(c) 30°, 50° (d) 30°, 40°

11 The opposite graph represents the alternating current through an electric circuit containing a hot-wire ammeter. So, the reading of the hot-wire ammeter equals the instantaneous value of current intensity after elapsing a time interval from the beginning of current cycle that equals ..... .



(a) 0.5 ms (b) 1 ms
(c) 2 ms (d) 4 ms

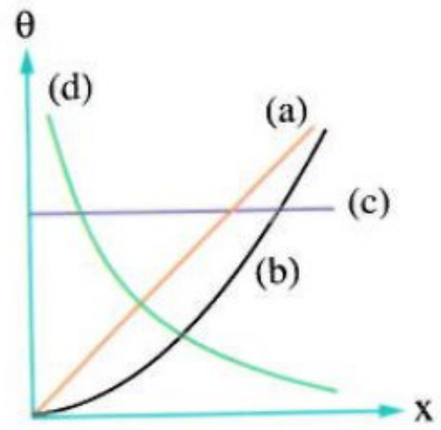
12 In the opposite electric circuit, when switch K is closed, an electric current of 1 A passes and the pointers of both ammeters deflect by the same angle. If the pointer of ammeter x deflects by an angle θ when an electric current of intensity 2 A passes in the circuit, then the pointer of ammeter y deflects by an angle that is ..... .



(a) less than θ (b) greater than θ
(c) equal to θ (d) the answer is indeterminable

Scanned with OKEN Scanner

13 The opposite graph represents the relationship between the deflection angle of a hot-wire ammeter's pointer (θ) and the quantity (x), so which of the shown curves (a), (b), (c) or (d) correctly represents this relationship if quantity x is:



(i) the electrical power consumed in the platinum-iridium wire?

(a) Curve (a) (b) Curve (b)

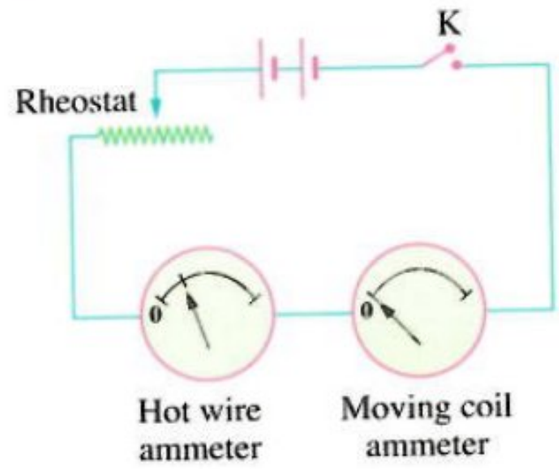
(c) Curve (c) (d) Curve (d)

(ii) the value of the electric current flowing in the platinum-iridium wire?

(a) Curve (a) (b) Curve (b)

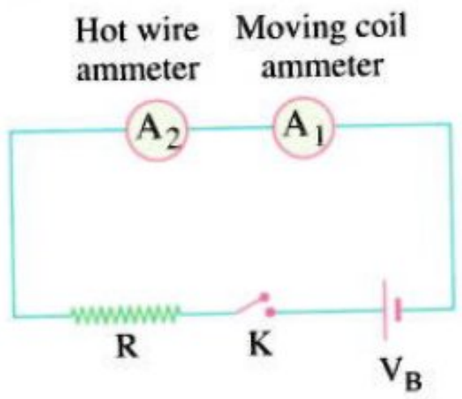
(c) Curve (c) (d) Curve (d)

14 The opposite electric circuit is used for calibrating a hot-wire ammeter. It was noticed that the pointer of the moving-coil ammeter points exactly to zero, while the pointer of the hot-wire ammeter points to a reading greater than zero. So, this means that the expansion coefficient of the plate on which the iridium-platinum wire is fixed is ..... the expansion coefficient of that wire.



(a) greater than (b) less than (c) equal to (d) indeterminable

15 A hot-wire ammeter and a moving coil ammeter are connected in an electrical circuit as shown in the opposite figure. When switch (K) is closed, which of the two ammeters will have its pointer settle first at a specific reading?



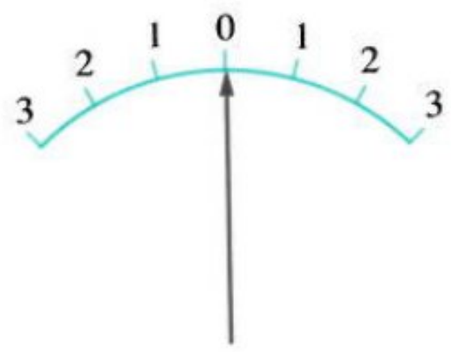
(a) The hot-wire ammeter.

(b) The moving coil ammeter.

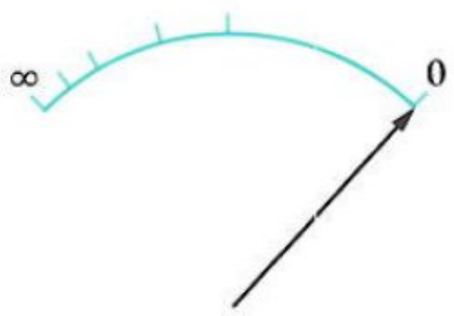
(c) The pointers of both ammeters settle together at the same time.

(d) The pointers of both ammeters do not move from the zero position.

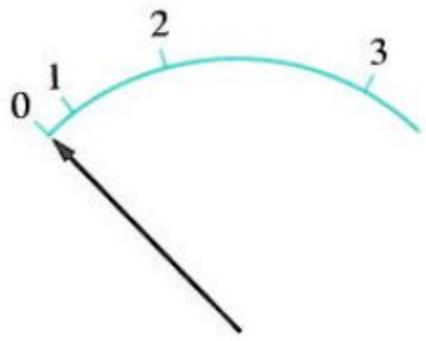
16 Each of the following figures illustrates the scale of one of three electrical measuring devices (1), (2) and (3):



(1)



(2)



(3)

CamScanner
Scanned with OKEN Scanner

So, which of these devices is used for:

(i) measuring the intensity of only direct current?

(a) Device (1) (b) Device (2)

(c) Device (3) (d) Both devices (1) and (3)

(ii) measuring the effective value of alternating current?

(a) Device (1) (b) Device (2)

(c) Device (3) (d) Both devices (1) and (3)

(iii) measuring an unknown ohmic resistance?

(a) Device (1) (b) Device (2)

(c) Device (3) (d) Both devices (1) and (3)

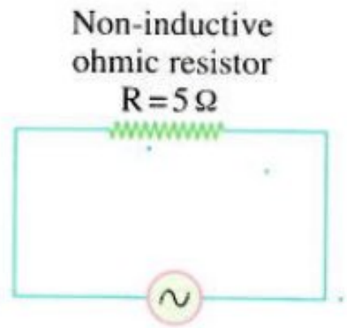
(iv) determining the direction of current in the circuit?

(a) Device (1) (b) Device (2)

(c) Device (3) (d) Both devices (1) and (3)

R circuit

17 In the opposite circuit, the alternating voltage across resistance R is ..... .



(a) in phase with the current

(b) leading the current by phase angle 90°

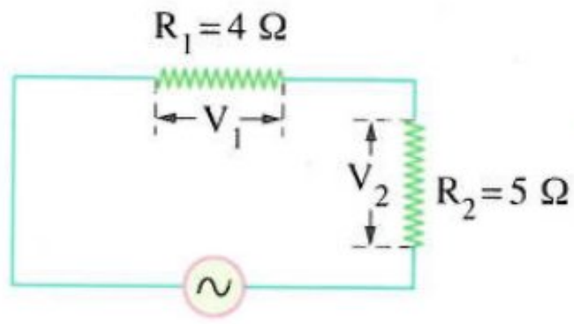
(c) lagging the current by $\frac{3}{4}$ cycle

(d) numerically equal to the current

18 An AC source is connected to an ohmic resistance of 100 Ω, if the instantaneous electromotive force of the source is given from the relation; V = 424.27 sin ωt, so the consumed power in the ohmic resistance is approximately equal to ..... .

(a) 760 W (b) 820 W (c) 850 W (d) 900 W

19 In the opposite circuit, the potential difference across resistor R_1 ..... the potential difference across resistor R_2



(a) is leading by a phase angle of 40°

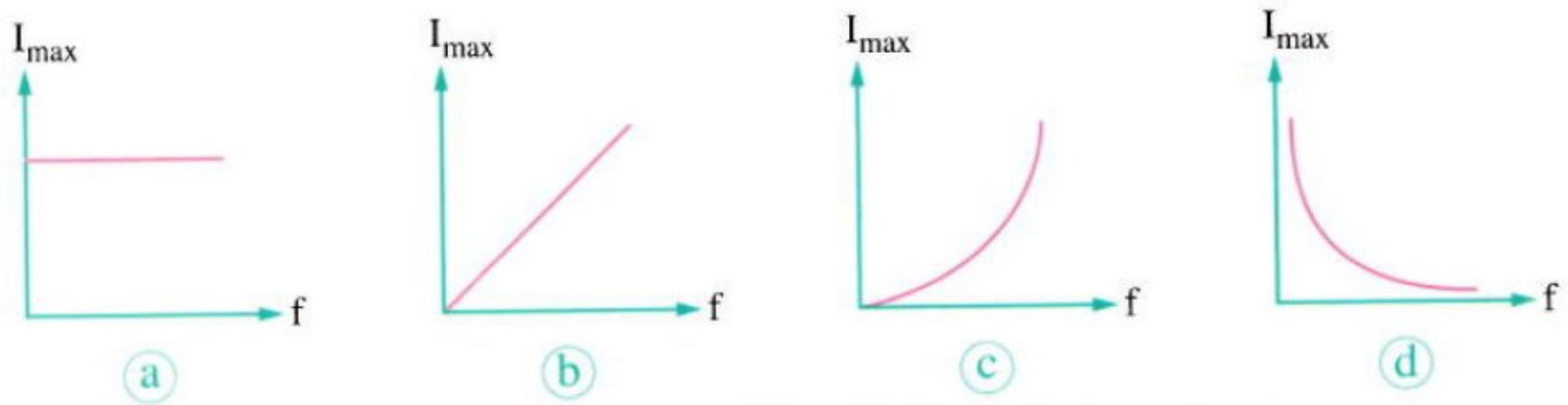
(b) is leading by a phase angle of 50°

(c) is lagging by a phase angle of 50°

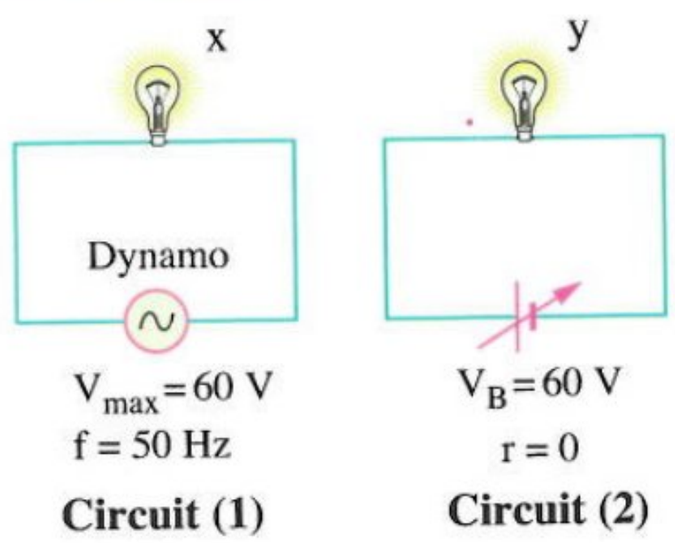
(d) is in phase with

Scanned with OKEN Scanner

20 Which of the following graphs represents the relation between the maximum value of the alternating current (I_{max}) through an ohmic resistance that is connected to an AC dynamo of negligible internal resistance and the frequency of the rotation of the dynamo's coil (f)?



21 Two identical lamps x and y. Lamp x is connected to a dynamo with a maximum voltage of 60 V and a frequency of 50 Hz as shown in circuit (1). Lamp y is connected to a DC source with an electromotive force of 60 V, negligible internal resistance, and changeable voltage as shown in circuit (2). In order for the brightness of the two lamps to be equal, we can ..... .



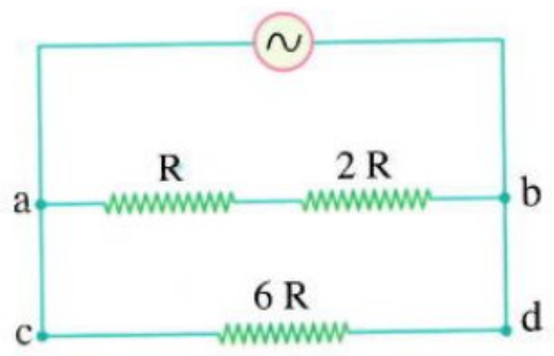
(a) increase V_B to $60\sqrt{2}$ V

(b) reduce V_B to 30 V

(c) increase the dynamo frequency to $50\sqrt{2}$ Hz

(d) reduce the dynamo frequency to $\frac{50}{\sqrt{2}}$ Hz

22 Three resistors with values R, 2 R, and 6 R are connected to an AC source as shown in the opposite electrical circuit. If the instantaneous value of the voltage across the source is given by the equation; $(V = 300\sqrt{2}\sin(100\pi t))$ and the effective value of the current flowing in branch cd equals 1 A, then:



(i) The instantaneous value of the total current flowing in the circuit is given by equation; .....

(a) $I = 3\sin(100\pi t)$

(b) $I = 3\sqrt{2}\sin(100\pi t)$

(c) $I = 3\sin(100\pi t + \frac{\pi}{2})$

(d) $I = 3\sqrt{2}\sin(100\pi t + \frac{\pi}{2})$

(ii) The value of resistance R equals ..... .

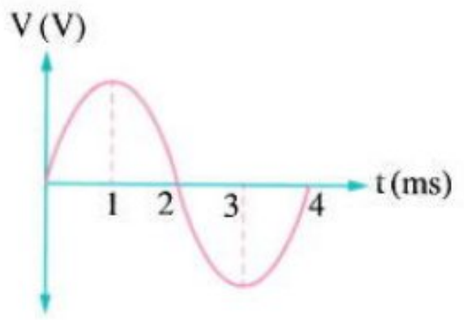
(a) 25 Ω (b) 50 Ω (c) 100 Ω (d) 200 Ω

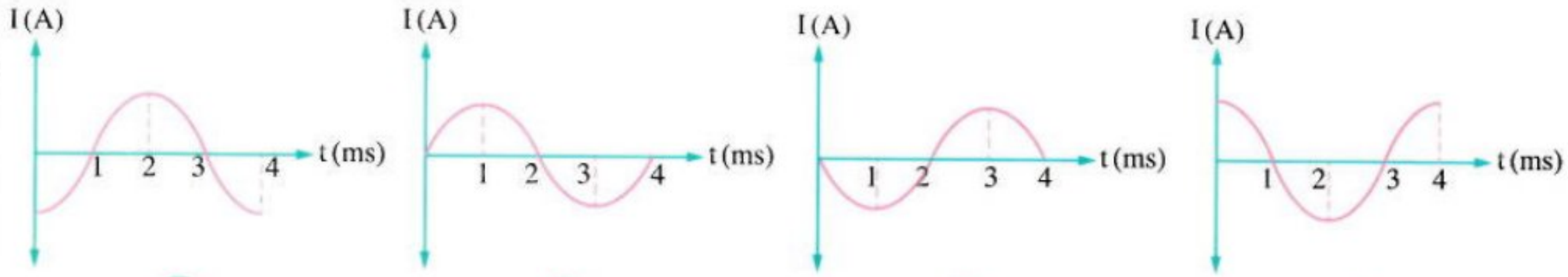
CamScanner

Scanned with OKEN Scanner

L circuit

23 If the potential difference (V) across the terminals of an inductor of negligible ohmic resistance which is connected to an AC source, is changing with time (t) as in the opposite graph, then the graph which represents the change in the current (I) through the inductor with the time (t) through the same period is ..... .





a b c d

24 If the inductive reactance of a coil which is connected to an AC source is (440 L) Ω where L is the self inductance of the coil, then the current frequency = ...

a 140 Hz b 400 Hz c 70 Hz d 44 Hz

25 An inductor has an inductive reactance of 1000 Ω. If each of the values of its self-inductance and the frequency of the current through it is doubled, its inductive reactance becomes ..... .

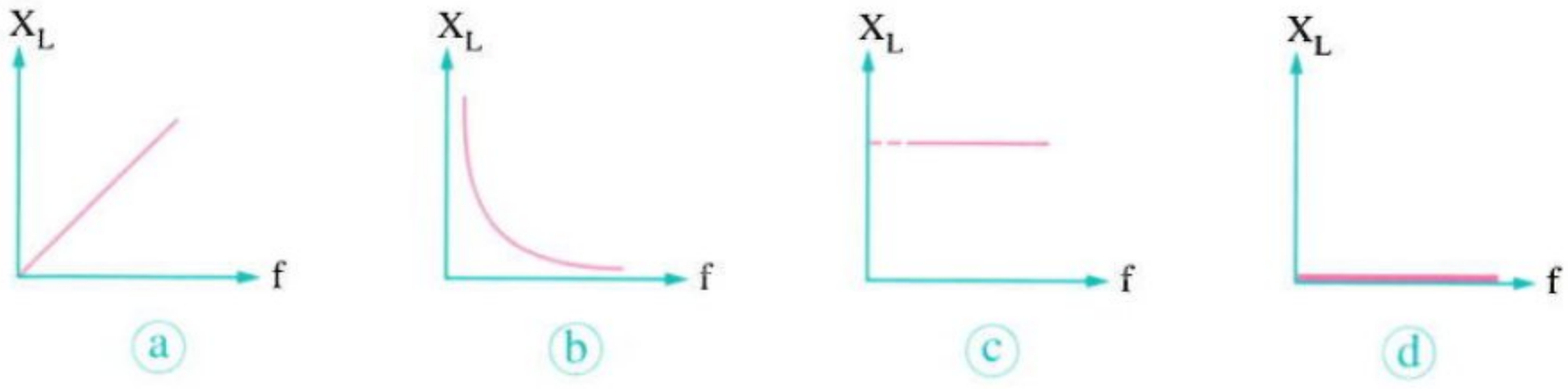
a 2000 Ω b 500 Ω c 250 Ω d 4000 Ω

26 AC current of effective value 100 mA is passing in an inductor of zero ohmic resistance and self-inductance 0.1 H. If the current frequency is 50 Hz, the effective potential difference across the inductor terminals equals ..... .

a 3.14 V b 31.4 V c 314 V d 3140 V

27 An AC dynamo with negligible internal resistance has its terminals connected to an inductor of negligible ohmic resistance. Which of the following graphs represents the relationship between:

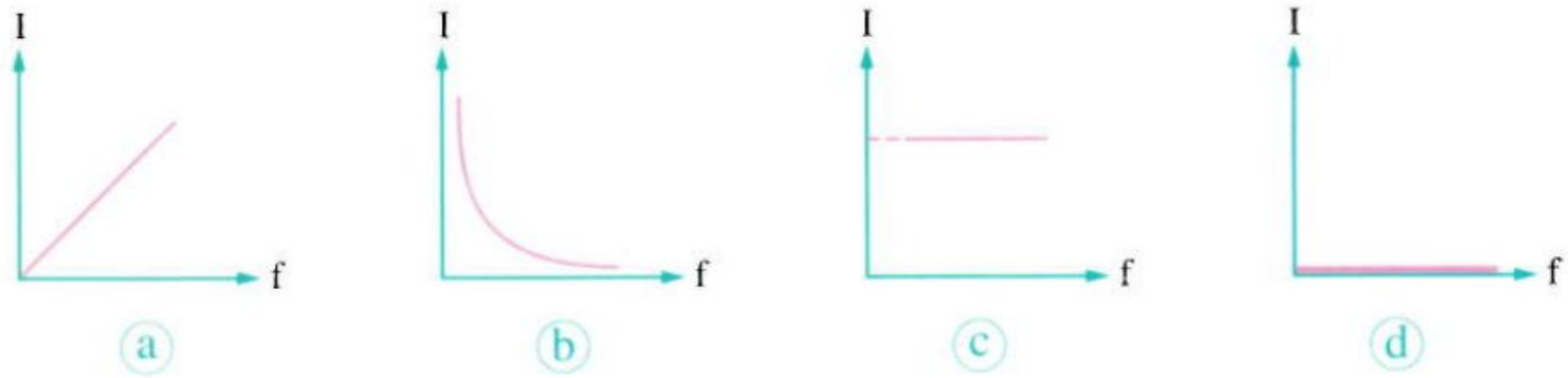
(i) the inductive reactance (X_L) of the inductor and the frequency (f) of the dynamo's rotation?



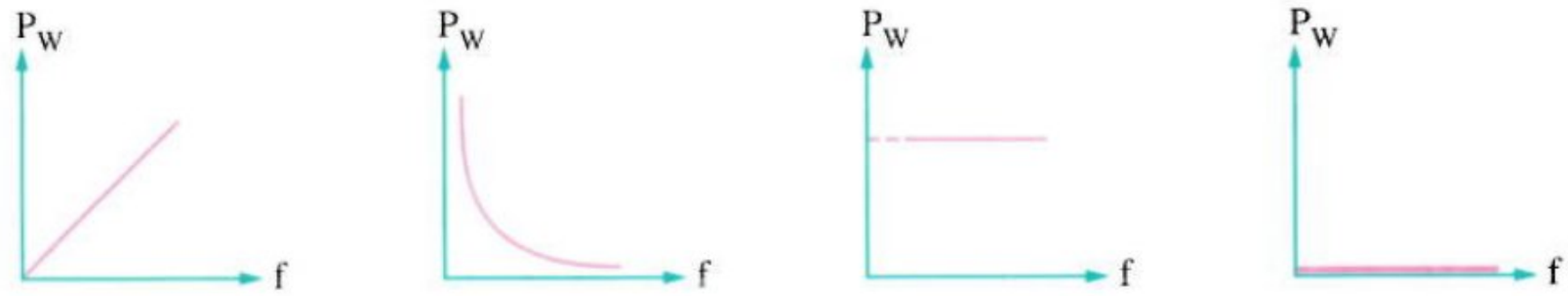
a b c d

Scanned with OKEN Scanner

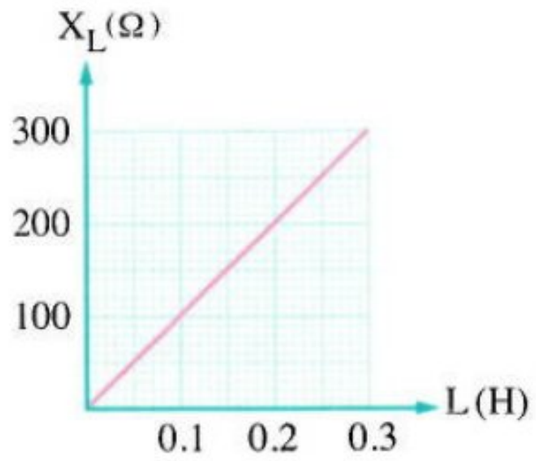
(ii) the value of the current (I) in the inductor and the frequency (f) of the dynamo's rotation?



(iii) the electrical power consumed in the inductor (P_w) and the frequency (f) of the dynamo's rotation?



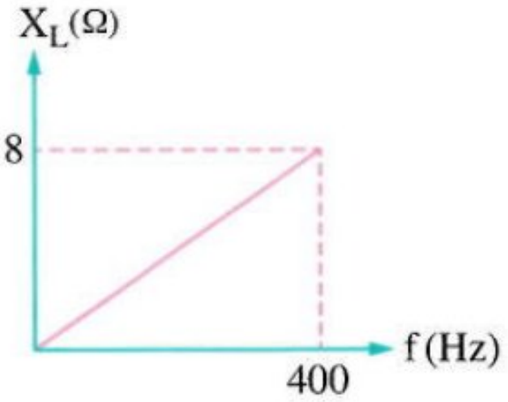
28 An inductor of zero resistance and changeable self inductance is connected to an AC source of frequency f and the opposite graph represents the relation between the inductive reactance of the inductor (X_L) and its self inductance (L), so the frequency (f) of the current equals ..... .



(a) 150.1 Hz (b) 159.1 Hz

(c) 162.1 Hz (d) 165.1 Hz

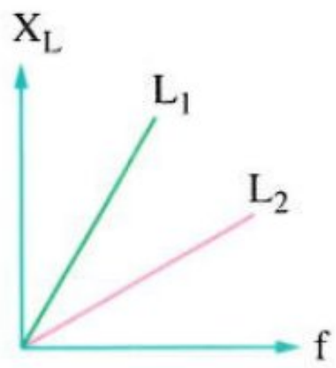
29 The opposite figure represents the graphical relation between the inductive reactance of a coil (X_L) and the frequency (f) of the current through it, then the self-inductance of the coil is ..... .



(a) 0.01 H (b) 0.02 H

(c) $\frac{1}{100\,\pi}$ H (d) $\frac{1}{50\,\pi}$ H

30 The opposite graph represents the relation between the inductive reactance (X_L) for two inductors whose self induction coefficients are L_1 and L_2 with respect to the frequency (f) of the AC current through each of them, so the relation between them is ..... .



(a) $L_1 = L_2 \neq 0$ (b) $L_1 > L_2$

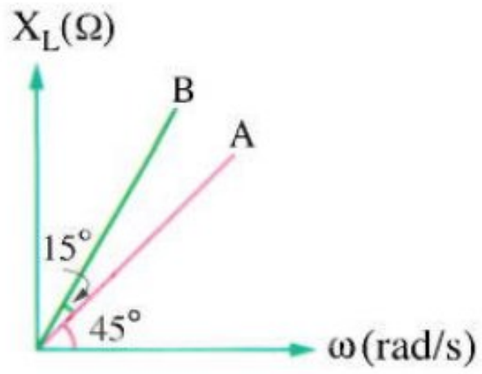
(c) $L_1 < L_2$ (d) $L_1 = L_2 = 0$

CamScanner

Scanned with OKEN Scanner

CamScanner

31 Two solenoids A and B are connected together in series to an AC dynamo of a changeable speed of rotation and the opposite graph represents the relation between the inductive reactance (X_L) for each solenoid and the angular speed (ω) of the dynamo's coil, so the ratio between the self-inductances of the two solenoids $\left(\frac{L_A}{L_B}\right)$ equals ..... .



(a) 0.02 (b) 0.15 (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{\sqrt{3}}{3}$

32 An inductor of zero ohmic resistance is connected to an AC source where the instantaneous potential difference across the terminals of the inductor is given by $V = 66 \sin(116\ \pi\ t)$, so if the maximum value of the current that passes in the circuit is 2 A, the self inductance of the inductor is approximately equal to ..... .

(a) 0.02 H (b) 0.05 H (c) 0.06 H (d) 0.09 H

33 An inductor is connected to an AC source so that the voltage across the inductor and the current flowing through it are given by the following equations:

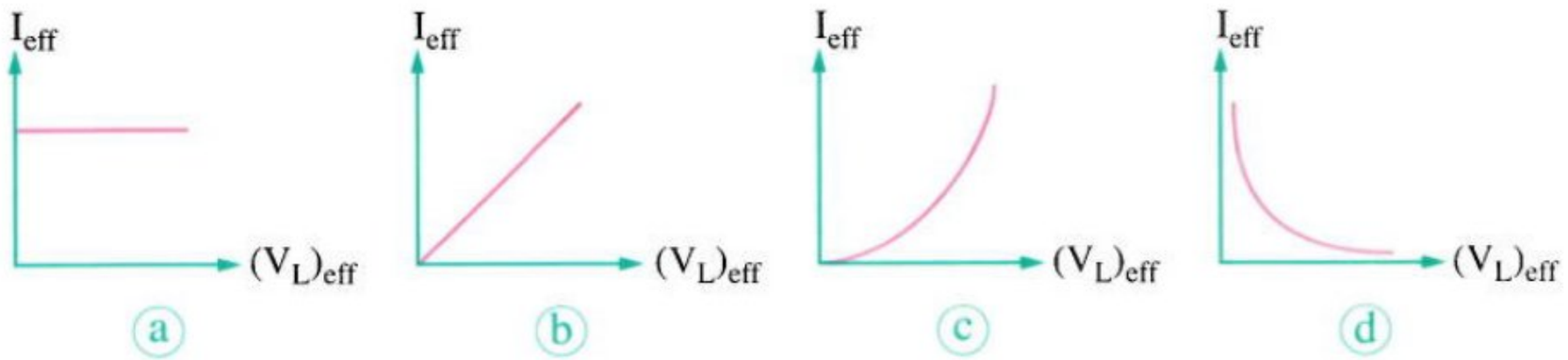
(1) $V = 200 \sin(50\ \pi\ t)$

(2) $I = 2 \sin(50\ \pi\ t - \frac{\pi}{2})$

So, the self-inductance of the inductor and the source frequency are ..... , ...

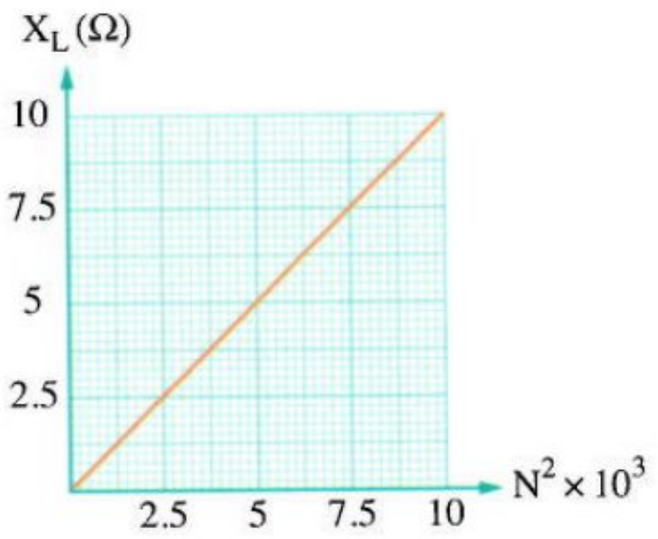
	Self-inductance of the inductor (H)	Source frequency (Hz)
(a)	$\frac{2}{\pi}$	25
(b)	$\frac{2}{\pi}$	50
(c)	π	25
(d)	π	50

34 A circuit contains an inductor of zero ohmic resistance connected to an AC source of constant frequency and changeable emf, so the graph that represents the relation between the effective value of AC current (I_{eff}) and the effective value of voltage across the terminals of the inductor $(V_L)_{eff}$ is ..... .



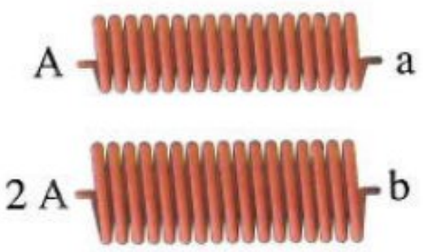
Scanned with OKEN Scanner

35 Several solenoids are different in the number of turns. Each has an area A_o and length 79 cm, a core magnetic permeability of 2×10^{-3} T.m/A and carrying an AC current of frequency 50 Hz. The opposite graph represents the relationship between the inductive reactance (X_L) for each of the solenoids and the square of their number of turns (N^2), so the area A_o of any of the solenoids equals ..... .



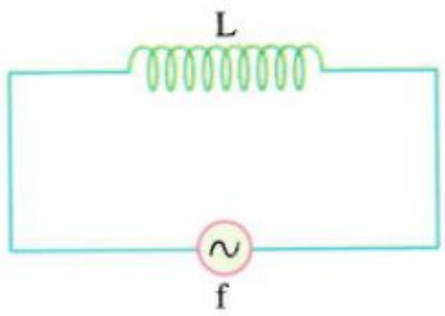
(a) 6.28 cm^2 (b) 12.6 cm^2 (c) 25.1 cm^2 (d) 62.8 cm^2

36 The opposite figure represents two solenoids (a and b) of the same length and the same number of turns, if the cross-sectional area of b is double that of a, so the ratios of their inductive reactances $\frac{(X_L)_a}{(X_L)_b}$ if they are connected to the same AC source equals ..... .



(a) $\frac{1}{2}$ (b) $\frac{3}{1}$ (c) $\frac{1}{1}$ (d) $\frac{2}{1}$

37 In the opposite figure, an inductor of self inductance L is connected to an AC source of frequency f, so its inductive reactance becomes X_L. If the inductor is cut into three identical parts and one of them is connected with the same electric source, then the self inductance and the inductive reactance of the inductor become ..... .



(a) $\frac{L}{3}, \frac{X_L}{3}$ (b) 3 L, $\frac{X_L}{3}$ (c) $\frac{L}{3}$, 3 X_L (d) 3 L, 3 X_L

38 An inductor has an air core and a self inductance L. The inductor is connected to an AC source of frequency f, so its inductive reactance becomes X_L, if a soft iron rod is inserted inside the inductor, then ..... .

	The self inductance of the inductor (L)	The inductive reactance of the inductor (X_L)
(a)	increases	decreases
(b)	decreases	increases
(c)	increases	increases
(d)	decreases	decreases

CamScanner
Scanned with OKEN Scanner

CamScanner

39 ✱ An inductor of zero ohmic resistance and self inductance 0.7 H is connected to an AC source of emf 120 V and frequency 50 Hz, then:

(i) The inductive reactance of the inductor is ..... .

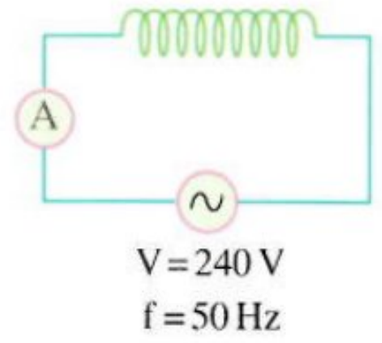
(a) 110 Ω (b) 200 Ω (c) 220 Ω (d) 300 Ω

(ii) The effective value of the current through the circuit is ..... .

(a) 0.4 A (b) 0.55 A (c) 0.6 A (d) 1.09 A

40 ✱ In the circuit shown in the figure, if the reading of the hot wire ammeter is 4 A, then the self inductance of the coil equals ..... .

(Knowing that the ohmic resistances of the coil and the ammeter are neglected)



(a) 0.191 H (b) 0.21 H (c) 0.251 H (d) 0.3 H

41 ✱ An AC source with a maximum voltage of 100 V had its terminals connected to an inductor with 132 turns, each with an area of 10^{-3} m^2, and a core magnetic permeability of 2×10^{-3} T.m/A, so the maximum value of the current flowing through it was 4 A lagging the maximum voltage by a time of 6.25 ms. So, the length of the coil equals ..... .

(a) 17.5 cm (b) 35 cm (c) 70 cm (d) 100 cm

42 ✱ When a current of frequency f_1 passes through a certain inductor of negligible ohmic resistance, the inductive reactance of the inductor becomes 15 Ω and when the current frequency is increased by 20 Hz, the inductive reactance of the inductor becomes 25 Ω, then the frequency of the current in the second case (f_2) equals ..... .

(a) 30 Hz (b) 40 Hz (c) 50 Hz (d) 60 Hz

43 ✱ When an electric current of frequency f passes through an inductor of zero ohmic resistance, its inductive reactance becomes 12 Ω and when the frequency gets increased by 20 Hz, its inductive reactance becomes 18 Ω, then:

(i) The ratio between the current frequency in the first case to that in the second case is ..... .

(a) $\frac{3}{2}$ (b) $\frac{2}{3}$ (c) $\frac{2}{1}$ (d) $\frac{1}{2}$

(ii) The self inductance of the coil is ..... .

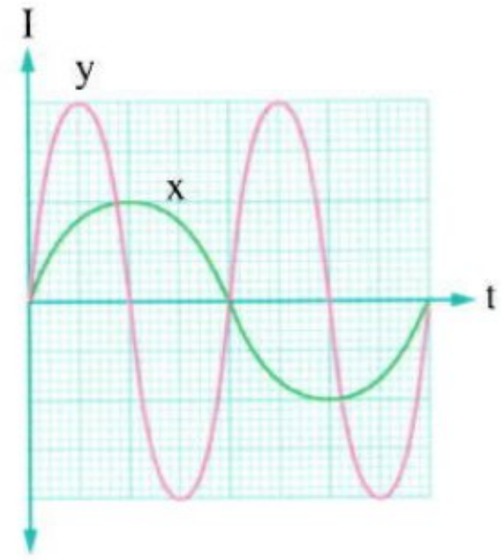
(a) 0.021 H (b) 0.031 H (c) 0.048 H (d) 0.071 H

Scanned with OKEN Scanner

44 ✱ A coil that consists of one layer of 300 turns, radius 2.1 cm and length 15 cm is wound around a cylinderical iron rod of permeability 0.002 Wb/A.m, then connected to an electric source of frequency 50 Hz, so the inductive reactance of the coil equals ..... .

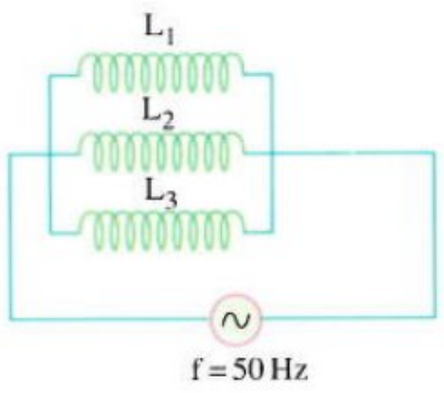
(a) 225.3 Ω (b) 341.4 Ω (c) 425.3 Ω (d) 522.7 Ω

45 Two AC sources, (1) and (2), have the same effective voltage. The terminals of source (1) are connected to an inductor x of negligible ohmic resistance, and the terminals of source (2) are connected to an inductor y of negligible ohmic resistance. The opposite graph represents the change in the value of the current (I) through the two inductors versus time (t), so the ratio between the self-inductance coefficients of the two inductors $\left(\frac{L_x}{L_y}\right)$ equals ..... .



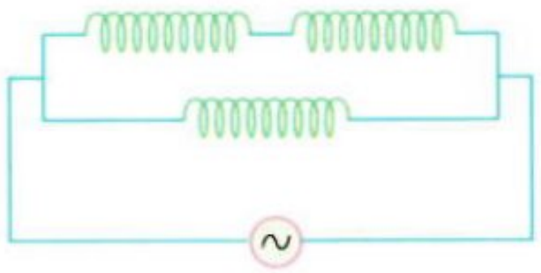
(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) 2 (d) 4

46 In the electric circuit shown in the figure, three coils L_1, L_2 and L_3 of zero resistance are separated away from each other and connected in parallel, they have inductances of 0.6 H, 0.2 H and 0.3 H, so the total network inductive reactance is ..... .



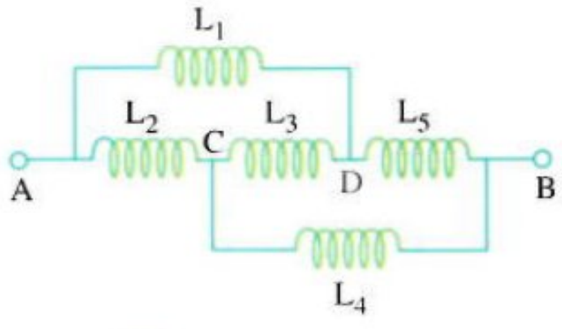
(a) 0.1 Ω (b) 6.28 Ω (c) 31.4 Ω (d) 100 Ω

47 In the electric circuit shown in the figure, if the coils are identical where the self-inductance for each is 0.3 H and the total inductive reactance is 12.56 Ω, then by ignoring the ohmic resistance and the mutual induction between the coils, the current frequency is ..... . (Where: $\pi = 3.14$)



(a) 50 Hz (b) 60 Hz (c) 20 Hz (d) 10 Hz

48 ✱ The opposite circuit consists of five inductors, each of self-inductance that equals 50 mH, so the total self-inductance between the two points A, B is ..... .

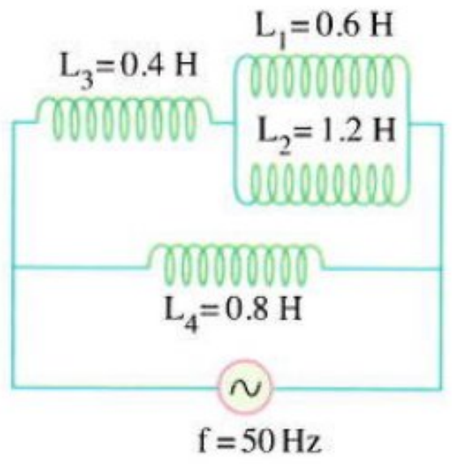


(a) 50 mH (b) 100 mH (c) 150 mH (d) 200 mH

CamScanner
Scanned with OKEN Scanner

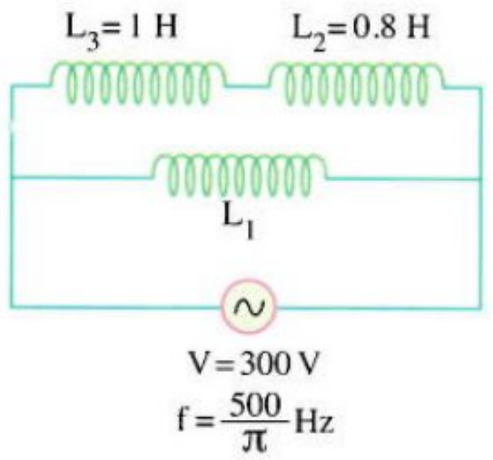
CamScanner

49 ✱ In the circuit shown in the figure,
the total inductive reactance is ..... .



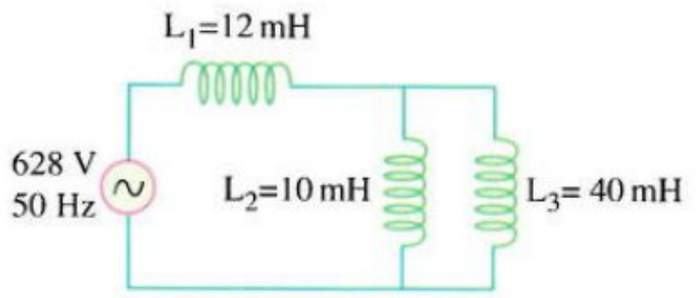
(a) 354.3 Ω
(b) 251.4 Ω
(c) 125.7 Ω
(d) 83.8 Ω

50 ✱ In the circuit shown in the figure:
If the electric current intensity passing
in the circuit is 0.5 A, then L_1 equals ..... .



(a) 0.6 H
(b) 0.9 H
(c) 1.8 H
(d) 1.9 H

51 ✱ The opposite circuit consists of a number of coils of zero ohmic resistance and an AC source, so the current intensities through each coil of the two coils L_2 and L_3 respectively are ..... .
(Where: $\pi = 3.14$)



(a) 20 A, 20 A (b) 80 A, 20 A (c) 80 A, 40 A (d) 80 A, 80 A

52 ✱ A group of identical inductors were connected together in a circuit carrying AC current of frequency 50 Hz. When they are connected in series their equivalent inductive reactance becomes 50 Ω and when they are connected in parallel it becomes 2 Ω, then:
(Ignoring the ohmic resistance of the inductors and the mutual induction between them)

(i) The number of inductors equals ..... .
(a) 10 (b) 6 (c) 5 (d) 4

(ii) The inductive reactance for each inductor is ..... .
(a) 5 Ω (b) 8 Ω (c) 10 Ω (d) 12.5 Ω

(iii) The self inductance of each of them approximately equals ..... .
(a) 0.039 H (b) 0.032 H (c) 0.018 H (d) 0.02 H

C circuit

53 A capacitor of capacitance 1 μF is connected to a DC source, so its capacitive reactance equals ..... .
(a) zero (b) ∞ (c) 1 Ω (d) $\frac{1}{2}$ Ω

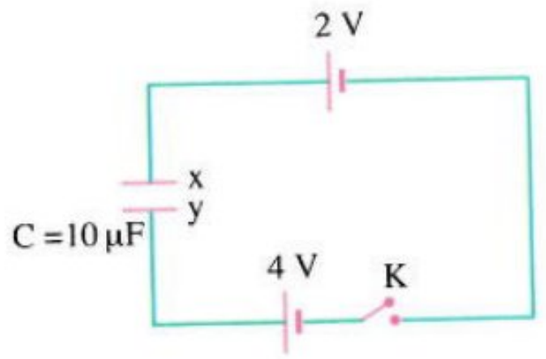
Scanned with OKEN Scanner

54 If the potential difference between the plates of a capacitor is 1 V and its capacitance is 3 μF, so the accumulated charge on one of its plates is ..... .

(a) 3 mC (b) 0.03 mC (c) 0.003 mC (d) 0.333 mC

55 In the following electric circuit, after closing switch K and the capacitor gets fully charged ..... .

(Where: $e = 1.6 \times 10^{-19}$C)



	Charge type on plate (y)	Number of electrons on plate (x)
(a)	negative	1.25×10^{14}
(b)	negative	2.5×10^{14}
(c)	positive	1.25×10^{14}
(d)	positive	2.5×10^{14}

56 The quantity $\sqrt{\frac{L}{C}}$ has the same measuring unit of ..... .

(Where: L is the self inductance of a coil and C is the capacitance of a capacitor)

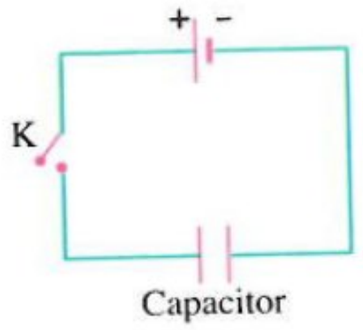
(a) resistance (b) time (c) frequency (d) potential difference

57 The quantity (CR) is measured in the unit of ..... .

(Where: C is the capacitance of the capacitor and R is the ohmic resistance)

(a) ohm (b) volt (c) second (d) hertz

58 In the opposite electric circuit, after closing switch K, the value of the electric current in the circuit ..... .



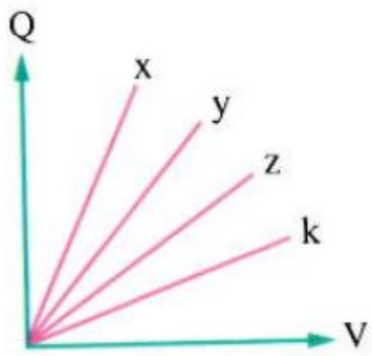
(a) increases as time passes

(b) decreases then increases

(c) vanishes when the capacitor is fully charged

(d) increases and decreases in the form of a sine wave curve

59 Four capacitors x, y, z and k are connected together in parallel to a DC source of changeable voltage and the opposite graph represents the relation between the accumulated charge (Q) on each of the four capacitors versus the potential difference (V) between the two plates of each of them, so which of them has the greatest capacitance?



(a) x (b) y (c) z (d) k

CamScanner

Scanned with OKEN Scanner

CamScanner

60 Figure (1) represents the graph of electric charge (Q) that is accumulated on one of the plates of each of three capacitors x, y and z versus potential difference (V) between the two plates of each of them, then if the three capacitors are connected as in figure (2), the equivalent capacitance between points a and b equals ..... .

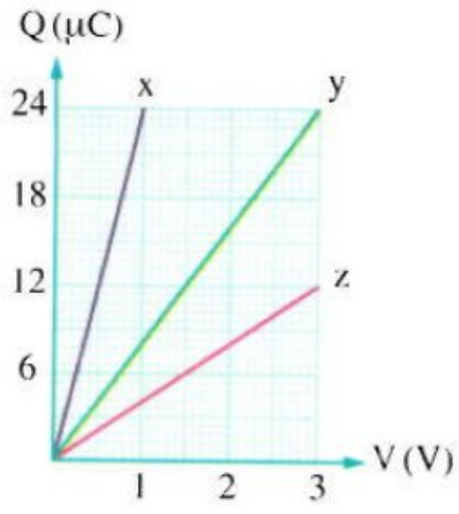


Figure (1)

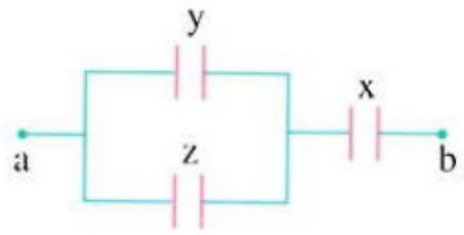


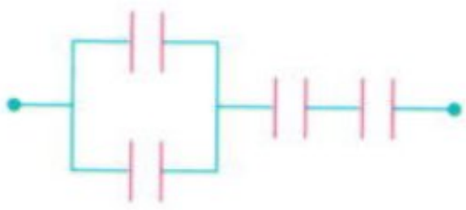
Figure (2)

(a) 6 μF (b) 8 μF (c) 10 F (d) 12 μF

61 Two capacitors have capacitances C_1 and C_2. When connected in series, their total capacitance equals 2 μF and when connected in parallel, their total capacitance becomes 9 μF. So, the capacitances of the two capacitors C_1 and C_2 are ..... , ...

(a) 4 μF, 5 μF (b) 3 μF, 6 μF (c) 4 μF, 4 μF (d) 2 μF, 9 μF

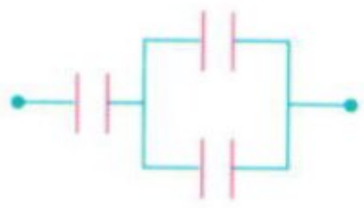
62 In the opposite figure, if the capacitance of each capacitor is 1 pF, then the total capacitance is ..... .



(a) 4 pF (b) 3 pF

(c) $\frac{1}{2}$ pF (d) $\frac{2}{5}$ pF

63 In the opposite figure, if the capacitance of each capacitor is C, then the total capacitance is ..... .

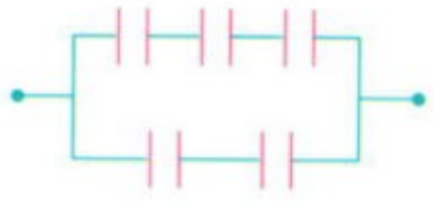


(a) 1.5 C (b) 3 C (c) $\frac{2}{3}$ C (d) C

64 Three capacitors of capacitances C_1, C_2 and C_3 are connected together in parallel and the group is connected between the terminals of a battery. If $C_3 > C_2 > C_1$ and the amounts of positive charge that are accumulated on each capacitor are Q_1, Q_2 and Q_3 respectively, then .....

(a) $Q_3 > Q_2 > Q_1$ (b) $Q_1 > Q_3 > Q_2$ (c) $Q_1 > Q_2 > Q_3$ (d) $Q_1 = Q_2 = Q_3$

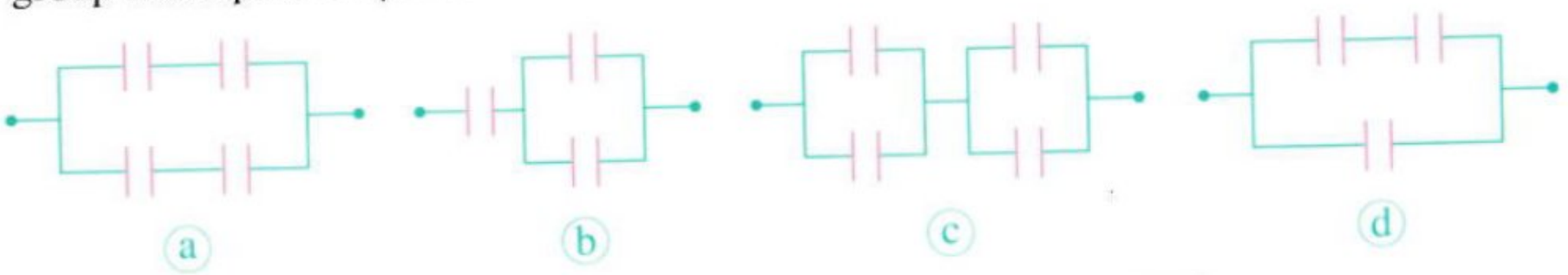
65 ✱ The opposite figure represents a set of capacitors, the capacitance of each is 24 pF, so the total capacitance of the group equals ..... .



(a) 4.8 pF (b) 16.8 pF (c) 20 pF (d) 24 pF

Scanned with OKEN Scanner

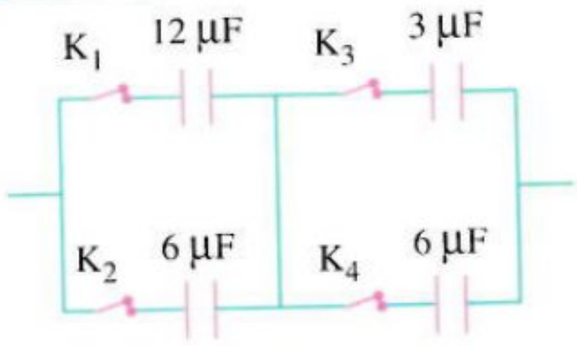
66 If you have a group of identical capacitors, the capacitance of each of them is 15 μF, then the method of connecting them together to get an equivalent capacitance for this group that equals 10 μF is ..... .



67 The total capacitance of a set of capacitors in an electric circuit is required to be decreased from 48 μF to 30 μF by adding a capacitor to this set, so the capacitance of the capacitor which is required to be added to the circuit and the way of its connection are ..... .

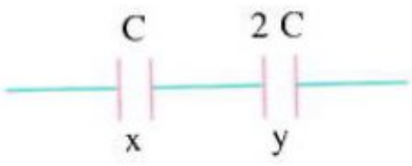
(a) 64 μF, in series
(b) 80 μF, in series
(c) 64 μF, in parallel
(d) 80 μF, in parallel

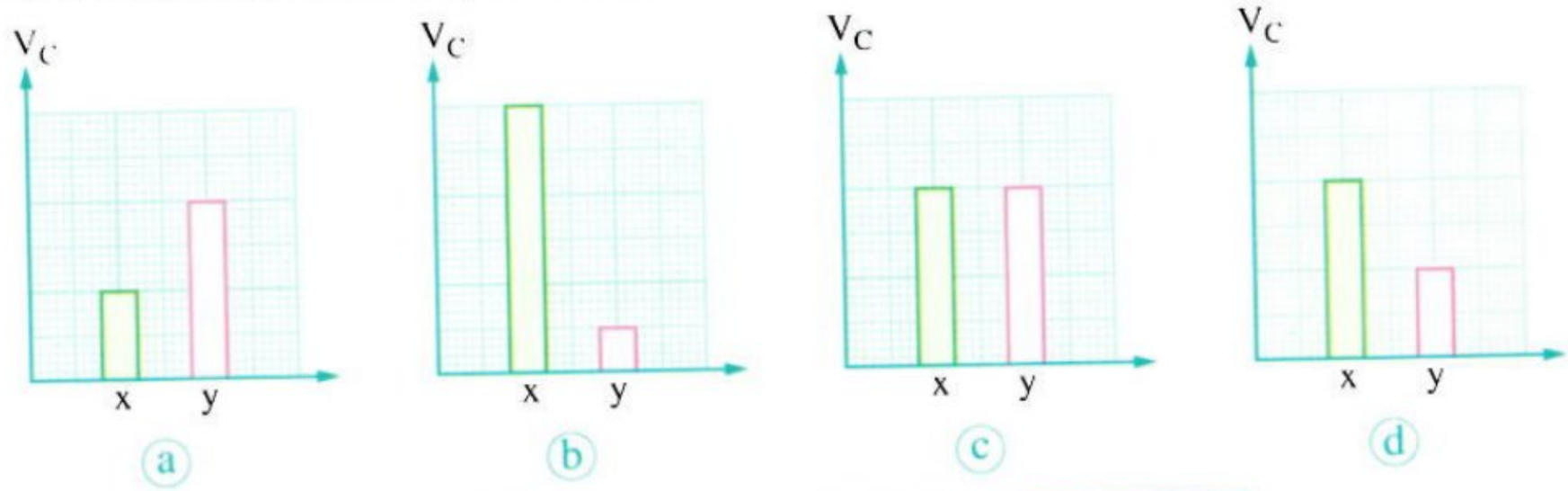
68 The opposite figure illustrates four capacitors connected together through four switches. Which of these switches, when opened, reduces the total capacitance by 1.5 μF?



(a) K_1
(b) K_2
(c) K_3
(d) K_4

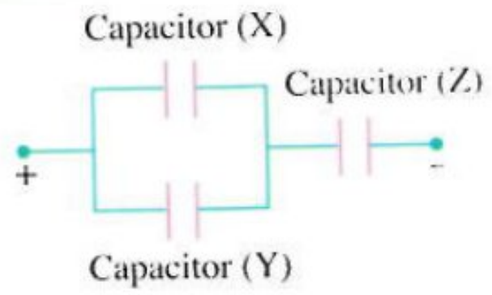
69 The opposite figure represents two capacitors x and y connected in series, which of the following graphs represents the proportions of potential differences between the two plates of each of them when they are connected to AC source?





(a) (b) (c) (d)

70 Three capacitors X, Y, and Z with capacitances C, 2 C, and $\frac{1}{2}$ C are connected together with a DC voltage as shown in the opposite figure, so the correct order of the three capacitors in terms of the charge accumulated on each of them is ..... .

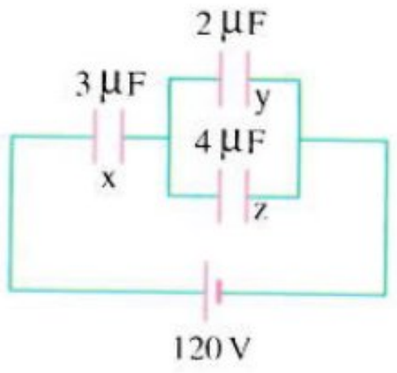


(a) $Z < X < Y$
(b) $Y < X < Z$
(c) $X < Y = Z$
(d) $X < Y < Z$

CamScanner
Scanned with OKEN Scanner

CamScanner

71 In the shown electric circuit, a DC source of voltage 120 V is connected to three capacitors x, y and z, then:



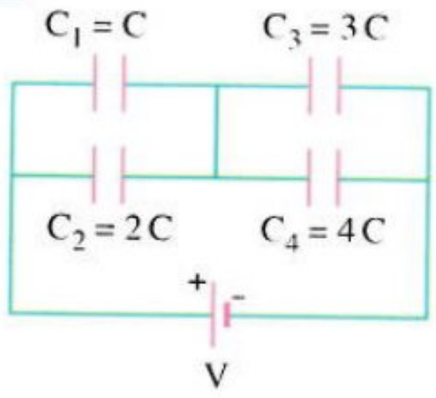
(i) The potential differences between the terminals of each of capacitors x and z respectively are ..... .

(a) 120 V, 60 V (b) 40 V, 80 V
(c) 80 V, 40 V (d) 60 V, 20 V

(ii) The accumulated positive electric charges on capacitors x and y respectively are ..... .

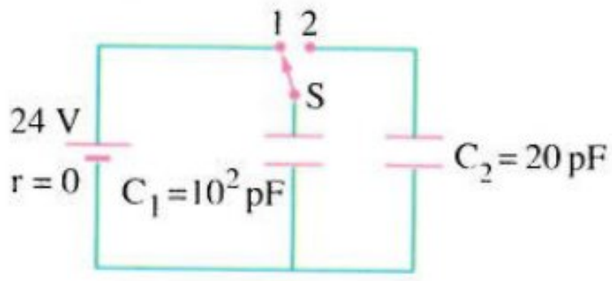
(a) 240 μC, 80 μC (b) 120 μC, 160 μC
(c) 80 μC, 120 μC (d) 160 μC, 120 μC

72 An electric circuit consists of many capacitors of different capacitances connected to an electric source of negligible internal resistance as in the figure. Which of these capacitors is charged with the largest charge?



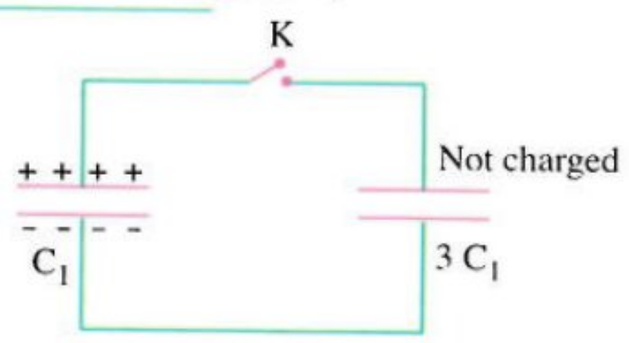
(a) C_1 (b) C_2
(c) C_3 (d) C_4

73 ✱ Two uncharged capacitors were connected to a battery of emf 24 V as shown in the opposite circuit, switch S was connected in position (1) until capacitor C_1 gets fully charged, so when the switch is connected in position (2), the potential difference between the terminals of capacitor C_1 becomes ..... .



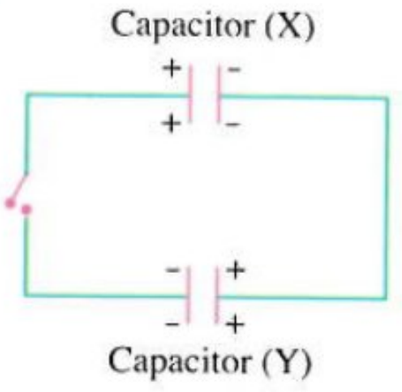
(a) 5 V (b) 10 V (c) 20 V (d) 15 V

74 A charged capacitor of capacitance C_1 is connected to an uncharged capacitor of capacitance $3\ C_1$ through a switch K as in the opposite figure, so when the switch K is closed, the charge of capacitor C_1 ..... .



(a) doubles (b) decreases to its half
(c) decreases to its one third (d) decreases to its one fourth

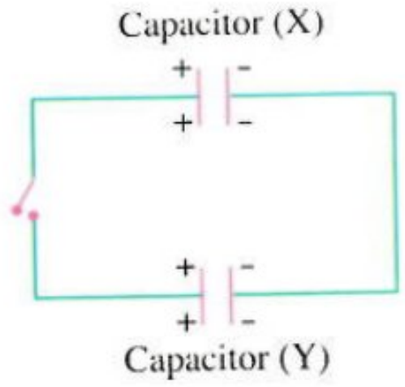
75 ✱ Two charged capacitors X and Y of capacitances C and 2C respectively have equal potential differences across them. The two capacitors are connected to each other through a switch (K) as shown in the opposite figure. What happens to the potential difference across each of the two capacitors (X and Y) respectively after closing the switch (K)?



(a) Doesn't change, doesn't change. (b) Decreases, decreases.
(c) Increases, decreases. (d) Decreases, increases.

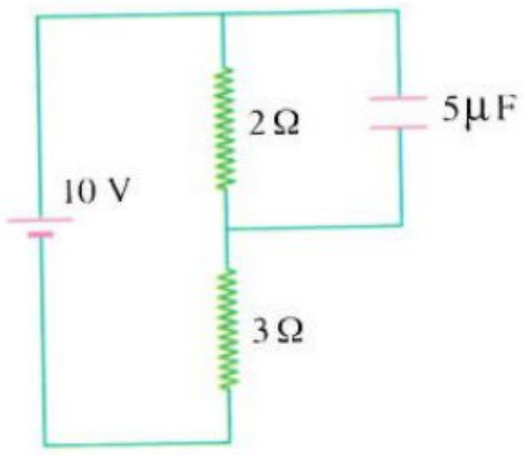
Scanned with OKEN Scanner

76 Two charged capacitors X and Y of capacitances C and 2C and potential differences across them 1 V and 2.5 V respectively are connected through a switch as shown in the opposite figure, so after closing the switch, the potential difference across capacitor X ..... .



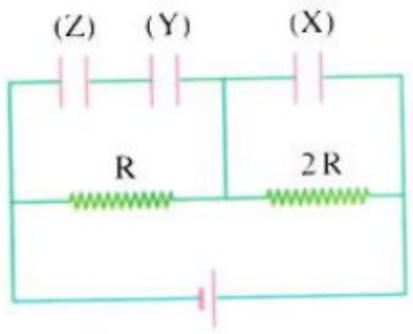
(a) remains 1 V
(b) becomes 2 V
(c) becomes 3 V
(d) becomes 4 V

77 In the opposite electric circuit, the accumulated positive charge on the capacitor is ..... .



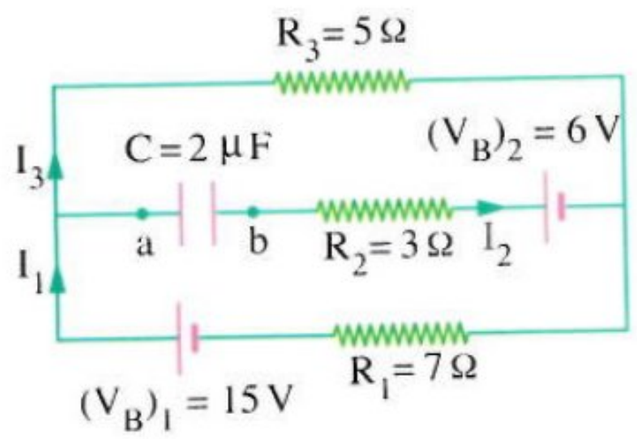
(a) 5 μC
(b) 10 μC
(c) 15 μC
(d) 20 μC

78 ✱ Three capacitors X, Y, and Z with capacitances C, 2 C and 4 C respectively are connected in an electric circuit as shown in the opposite figure, so the correct order of the capacitors according to the voltage across each of them is .....



(a) $V_Z < V_Y < V_X$
(b) $V_x < V_Y < V_z$
(c) $V_X = V_Y = V_Z$
(d) $V_Y < V_Z < V_X$

79 ✱ In the electric circuit shown in the figure;



(i) The value of each I_1, I_3 respectively equals ..... .

(a) 3 A, 2 A
(b) 0.75 A, 0.75 A
(c) 1.25 A, 0.5 A
(d) 1.25 A, 1.25 A

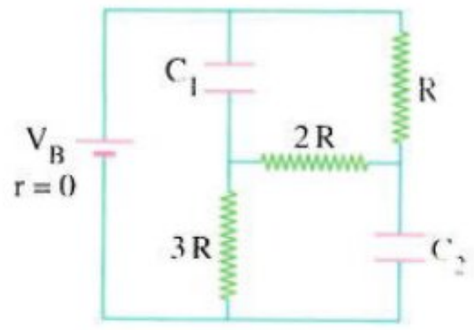
(ii) The positive accumulated charge on one of the plates of the capacitor equals ..... .

(a) 0.125 μC
(b) 0.5 μC
(c) 3.2 μC
(d) 5.5 μC

CamScanner
Scanned with OKEN Scanner

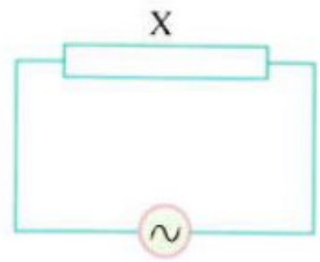
CamScanner

80 The opposite electric circuit contains two charged capacitors C_1 and C_2. If their capacitances are C and 2 C respectively, then the ratio between the quantities of electric charges on the two capacitors $\left(\frac{Q_1}{Q_2}\right)$ is ..... .



(a) $\frac{1}{2}$ (b) $\frac{3}{5}$ (c) $\frac{1}{1}$ (d) $\frac{3}{10}$

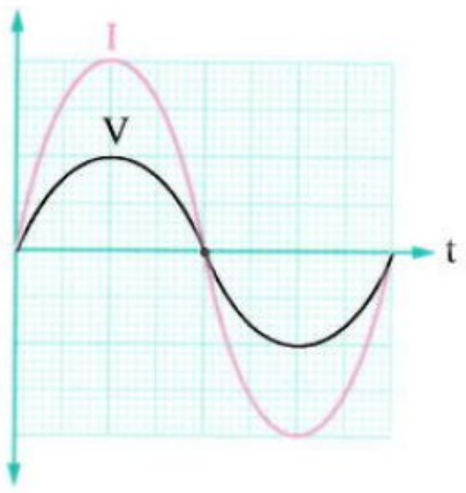
81 In the circuit shown in the figure, if the voltage lags the current, then component (x) is ..... .



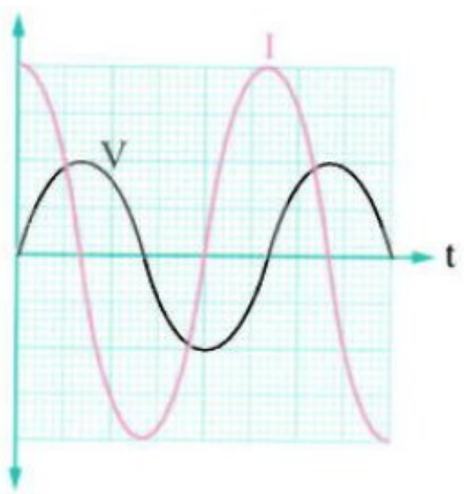
(a) a capacitor (b) an ohmic resistance

(c) an inductor of negligible ohmic resistance (d) a hot wire ammeter

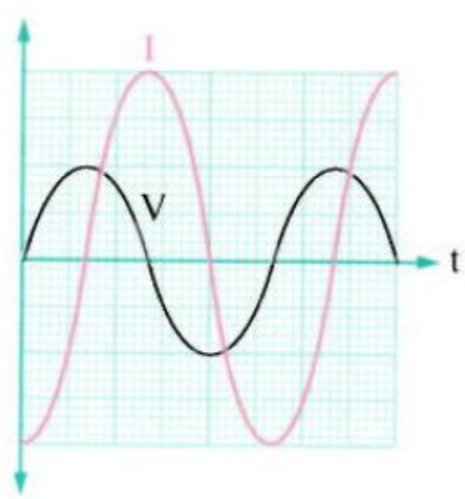
82 An AC source is connected to three electrical components X, Y, and Z, each separately. The following graphs represent the change in both the potential difference across the source (V) and the value of current (I) flowing through it versus time (t):



The state of component (X)



The state of component (Y)



The state of component (Z)

So, the electrical components X, Y, and Z are:

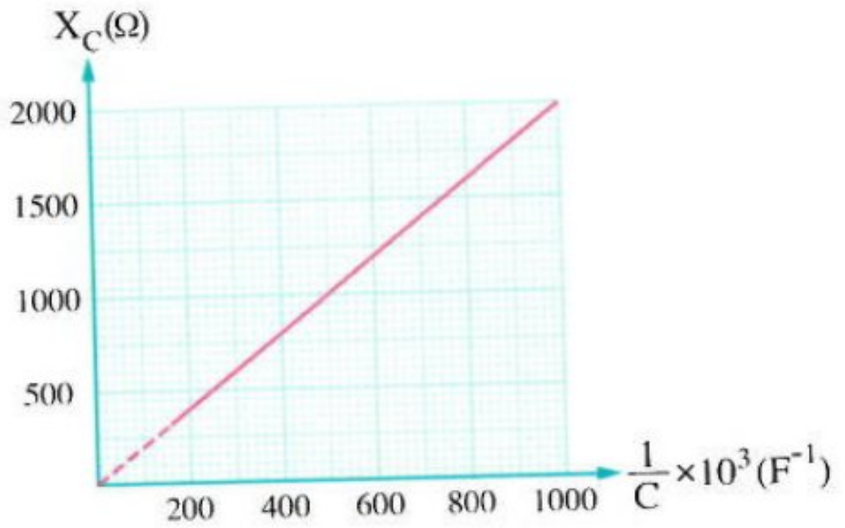
	Component X	Component Y	Component Z
(a)	Resistance without inductance	Capacitor	Inductor without resistance
(b)	Resistance without inductance	Inductor without resistance	Capacitor
(c)	Inductor without resistance	Capacitor	Resistance without inductance
(d)	Inductor without resistance	Resistance without inductance	Capacitor

Scanned with OKEN Scanner

83 A capacitor of constant capacitance is directly connected to an AC source of frequency f so that it has a capacitive reactance X_C. If the current frequency is tripled, the capacitive reactance ..... .

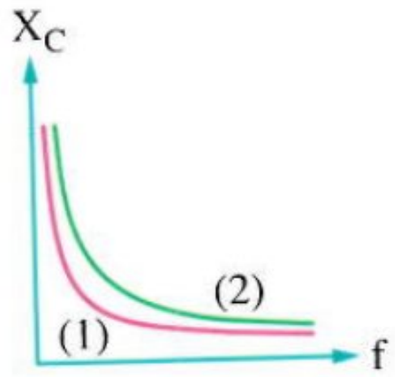
(a) triples
(b) becomes nine times as much
(c) decreases to its third
(d) remains unchanged

84 The opposite graph represents the relation between the capacitive reactance (X_C) of a capacitor which has a changeable capacitance and the reciprocal of its capacitance ($\frac{1}{C}$) when it is connected to an AC source of frequency f, so the frequency of the current (f) equals ..... .



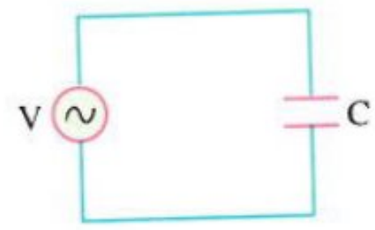
(a) 52.55 Hz
(b) 64.54 Hz
(c) 79.55 Hz
(d) 81.48 Hz

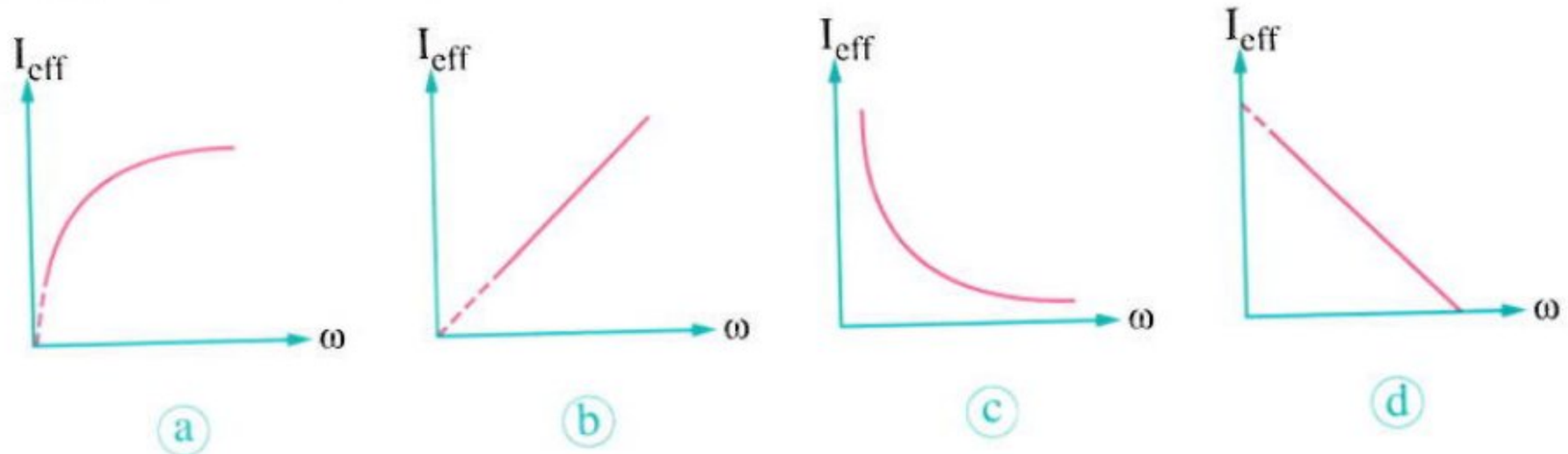
85 * The opposite graph represents the relation between the capacitive reactance (X_C) and the frequency (f) for two capacitors of capacitances C_1 and C_2, so the relation between them is ..... .



(a) $C_1 = C_2$
(b) $C_1 > C_2$
(c) $C_1 < C_2$
(d) indeterminable

86 AC source of changeable frequency and constant effective value of voltage is connected to a capacitor of capacitance C as shown in the figure, which of the following graphs represents the relation between the effective value of current intensity in the circuit (I_{eff}) and the angular frequency (ω) of the source?

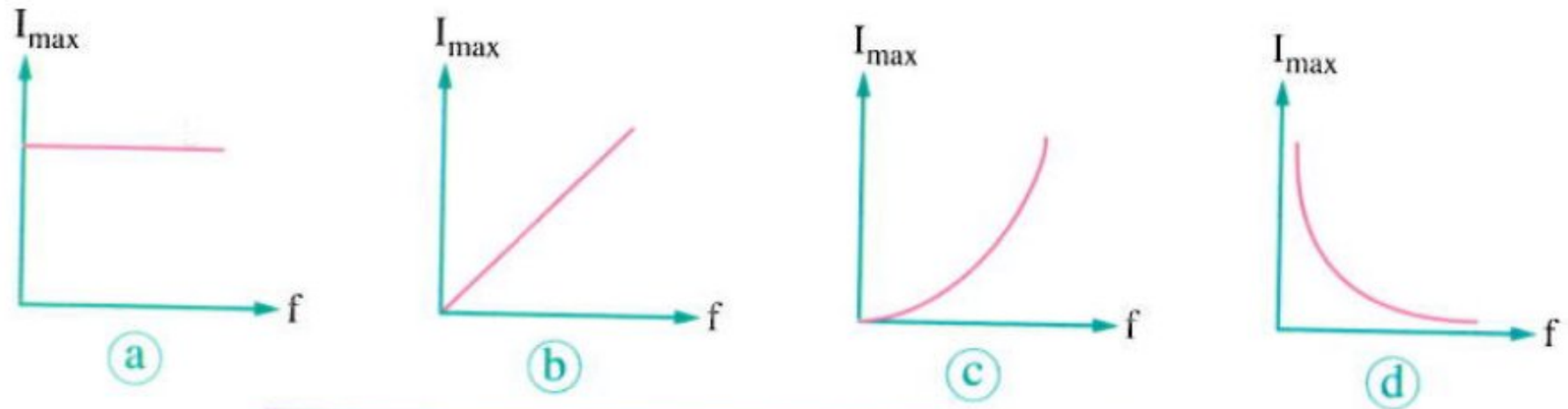




CamScanner
Scanned with OKEN Scanner

CamScanner

87 A circuit contains an AC dynamo of negligible internal resistance connected to a capacitor, so the graph which represents the relation between the maximum value of the AC current (I_{max}) through the capacitor's circuit and the frequency of rotation (f) of the dynamo's coil is ..... .



88 ✱ A capacitor of capacitance $\frac{7000}{11}$ μF is connected to an AC source of 20 V and frequency 50 Hz, then:

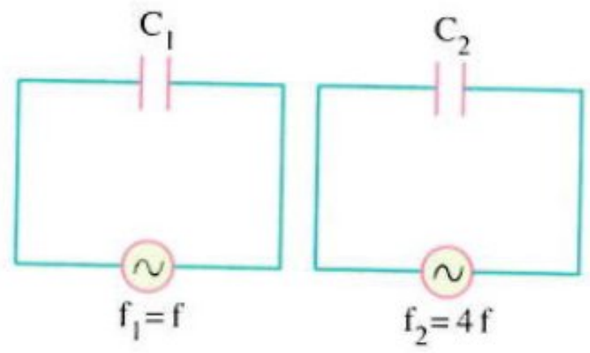
(i) The capacitive reactance of a capacitor equals ..... .

(a) 0.1 Ω (b) 0.2 Ω (c) 5 Ω (d) 10 Ω

(ii) The value of the electric current passing in the circuit is ..... .

(a) 1 A (b) 2 A (c) 3 A (d) 4 A

89 The opposite figure shows two circuits, each contains an AC source and a capacitor, if: $\frac{(X_C)_1}{(X_C)_2} = \frac{2}{3}$, then .....



(a) $\frac{C_1}{C_2} = \frac{3}{4}$ (b) $\frac{C_1}{C_2} = \frac{6}{1}$

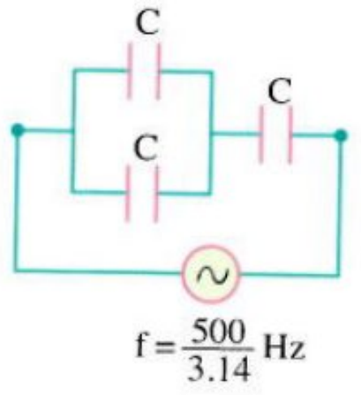
(c) $\frac{C_1}{C_2} = \frac{8}{3}$ (d) $\frac{C_1}{C_2} = \frac{1}{12}$

90 A capacitor of capacitance $\frac{700}{22}$ μF is connected to an AC source whose instantaneous voltage is given by the equation; ($V = 100 \sin(200 \pi t)$), so the instantaneous value of the current flowing in the circuit is given by the equation; .....

(a) $I = 2 \sin(200 \pi t)$ (b) $I = 0.5 \sin(200 \pi t + \frac{\pi}{2})$

(c) $I = 2 \sin(200 \pi t - \frac{\pi}{2})$ (d) $I = 2 \sin(200 \pi t + \frac{\pi}{2})$

91 In the opposite figure, if all capacitors have equal capacitances and their total capacitive reactance is 50 Ω, then the capacitance of each capacitor (C) equals ..... . (Where: $\pi = 3.14$)

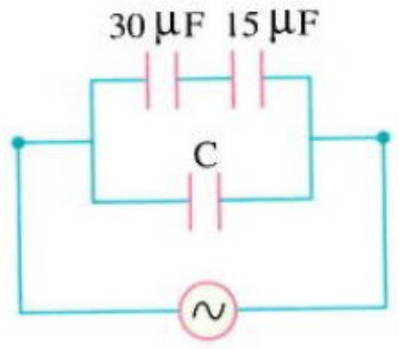


(a) 2 μF (b) 6 μF

(c) 12 μF (d) 30 μF

Scanned with OKEN Scanner

92 AC source of effective voltage 500 V and frequency $\frac{100}{\pi}$ Hz is connected to a set of capacitors as shown in the opposite figure, so an electric current of effective value 2 A passes through the circuit, then the capacitance of capacitor C equals ..... .



(a) 15 μF (b) 10 μF
(c) 20 μF (d) 50 μF

---

93 ✱ Three capacitors, the capacitance of each is 14 μF, are connected together in parallel and with an AC source of frequency 50 Hz, then the total capacitive reactance equals ..... .

(a) 681.8 Ω (b) 322.3 Ω (c) 151.5 Ω (d) 75.76 Ω

---

94 ✱ Three capacitors of capacitances 10 μF, 20 μF, 30 μF are connected in series with an electric source of emf 200 V and frequency 42 Hz, then:

(i) The total capacitive reactance equals ..... .

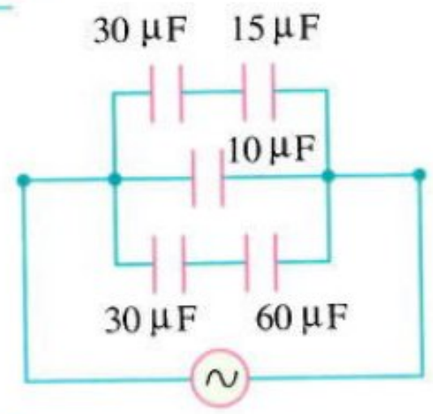
(a) 1388.8 Ω (b) 695.02 Ω (c) 126.26 Ω (d) 63.13 Ω

(ii) The value of the current passing in the circuit is ..... .

(a) 3.16 A (b) 1.58 A (c) 0.29 A (d) 0.14 A

---

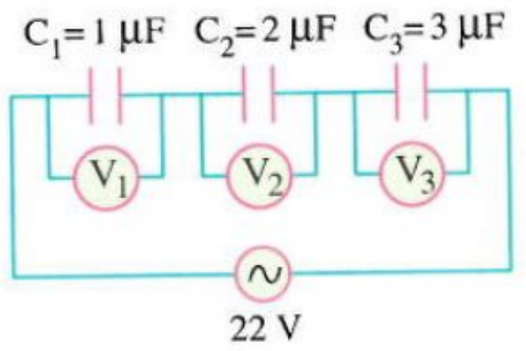
95 ✱ An AC source of voltage 400 V and frequency 50 Hz is connected to a group of capacitors as shown in the opposite figure, the total current intensity passing in the electric source is ..... .



(a) 0.94 A (b) 2.03 A
(c) 4.02 A (d) 5.03 A

---

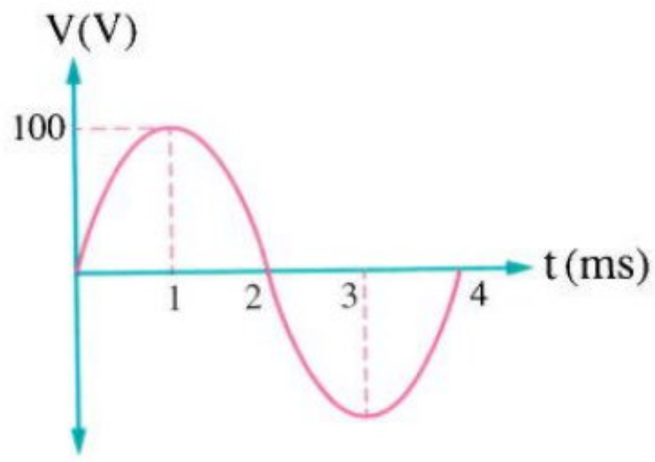
96 ✱ From the opposite figure, ...



	V_1	V_2	V_3
(a)	12 V	5 V	5 V
(b)	10 V	8 V	4 V
(c)	12 V	6 V	4 V
(d)	4 V	6 V	12 V

---

97 ✱ The opposite graph represents the change of the emf (V) of a dynamo's coil with time (t), if this dynamo is connected to a capacitor of capacitance 2 μF. Then the effective value of the current that passes in the source is ..... .



(a) 0.445 A (b) 0.314 A
(c) 0.22 A (d) 0.15 A

CamScanner

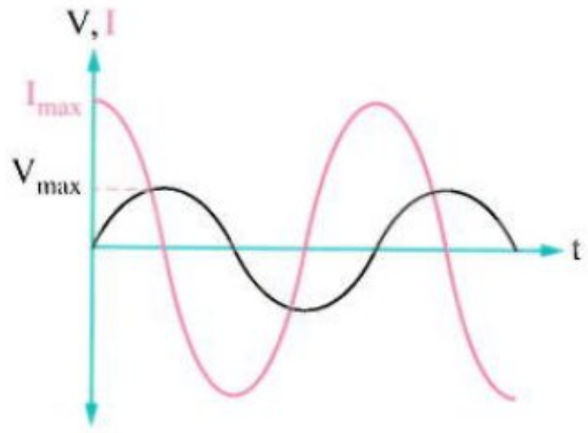
Scanned with OKEN Scanner

CamScanner

98 A dynamo with 100 turns and an area of 0.05 m^2 is subjected to a uniform magnetic field with a flux density of 0.225 T. The ends of this coil are connected to a capacitor, resulting in a capacitive reactance of 1 kΩ and an effective current of 0.25 A through the circuit. So, the angular frequency of the dynamo is approximately ..... .

(a) 157 rad/s (b) 314 rad/s (c) 471 rad/s (d) 628 rad/s

99 An AC source with negligible internal resistance is connected with a pure component such that the opposite graph represents the variation of the current intensity (I) through the circuit and the voltage (V) across the source versus time (t), so the electric power consumption through the circuit equals ..... .



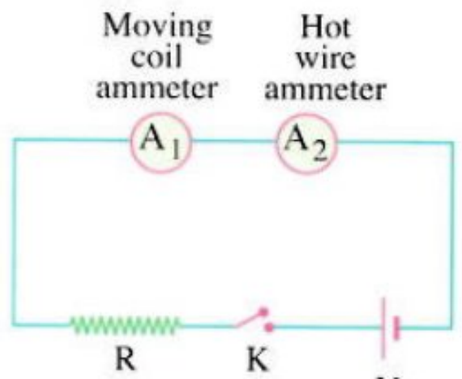
(a) $I_{max} V_{max}$ (b) $\frac{I_{max} V_{max}}{2}$ (c) $\frac{I_{max} V_{max}}{\sqrt{2}}$ (d) zero

Second Essay questions

Hot-wire ammeter

1 **Give reasons for:** Measuring the value of AC current depends on its thermal effect.

2 The opposite figure shows an electric circuit having two ammeters A_1, A_2 of different types, at closing switch K:



(a) **What** is the type of energy to which the electrical energy gets converted causing the pointer to deflect in each of the two devices?

(b) **In which** of the ammeter devices the pointer settles firstly at a specific reading? **And why**?

(c) **In which of the two devices:**

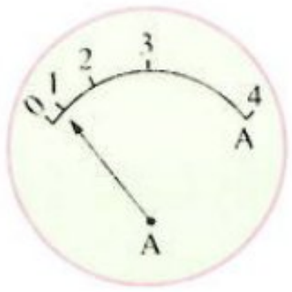
(1) an electric current passes through the spring coil?

(2) the spring coil deflects the pointer from the zero position?

3 When using a hot-wire ammeter in measuring a DC current intensity, **does** that require calibrating the ammeter scale to be uniform before starting the measuring process? **And why**?

Scanned with OKEN Scanner

4 A student wanted to use a hot wire ammeter to measure the current in an electric circuit but he noticed that the pointer of the ammeter settles at reading 1 A before passing an electric current through it as shown in the opposite figure:



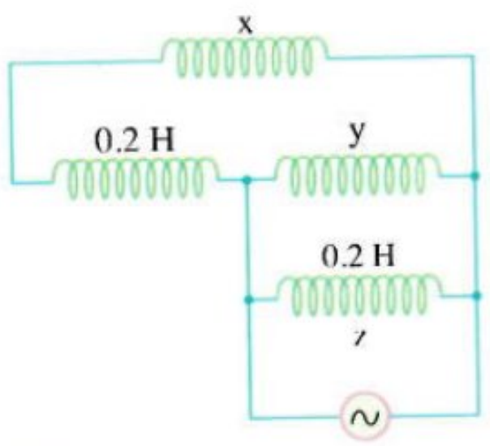
(a) **What is** the possible reason for not settling the pointer at zero even though no current is flowing through the ammeter?

(b) The student thought of passing the electric current in the circuit and recording the reading at which the pointer settles, then subtracting 1 A from it to correct the zero error. **Is this idea scientifically accurate? And why?**

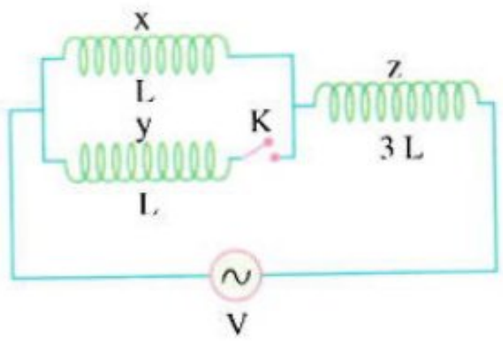
(c) **What** solutions do you suggest to avoid not having the pointer settled at zero before passing the current?

L circuit

5 In the electric circuit shown in the figure, if all the coils have negligible ohmic resistances and the value of the current through coil z is double the current through coil x and half the current through coil y, **calculate** the self inductances of the two coils x, y.



6 AC circuit consists of three coils of negligible ohmic resistance, AC source and switch K as shown in the opposite figure, **what happens when the switch (K) is closed for each of:**

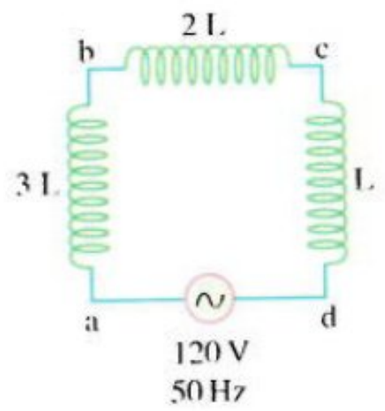


(a) The phase difference between the total voltage and the current in the circuit?

(b) The voltage across the terminals of coil z?

(c) The value of the current through coil x?

7 In the opposite electric circuit, if the coils have a negligible ohmic resistances:



(a) **Find** the potential difference between the two points:

① a, c ② b, d

(b) If the value of the current passing through the circuit is 1 A, **find** the value of inductance L

8 **What** are the results if the wires of a coil that carries an AC current are double folded before being coiled concerning the inductive reactance of the coil?

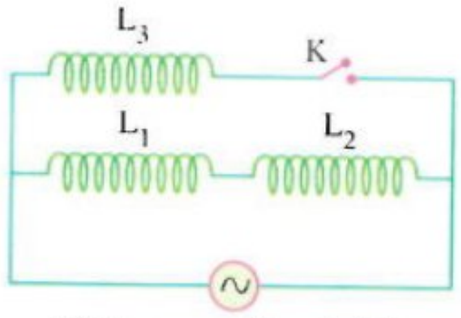
CamScanner

Scanned with OKEN Scanner

CamScanner

9 **Prove that** the quantity $\frac{L}{R}$ has the same measuring unit of time, where L is the self-inductance of a coil and R is the ohmic resistance.

10 **In the opposite electric circuit, what happens to the maximum value of the current that passes through the coil L_1 if the coils have negligible ohmic resistances when:**

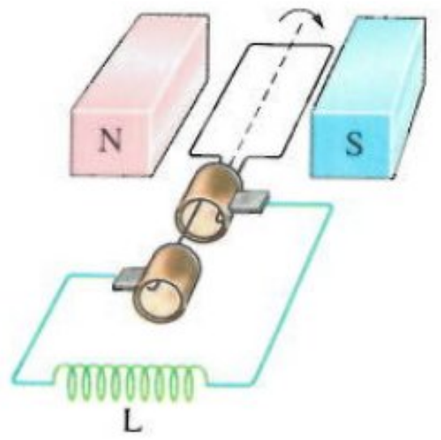


AC dynamo of negligible ohmic resistance

(a) closing switch K?

(b) increasing the frequency of the source?

11 The opposite figure shows an AC dynamo of negligible ohmic resistance whose terminals are connected to an inductor (L) of negligible ohmic resistance, **what will happen** to the value of the current in the dynamo's coil when:



(a) bringing the turns of the coil (L) closer to each other regularly (uniformly)?

(b) connecting another coil (L) in series in the external circuit?

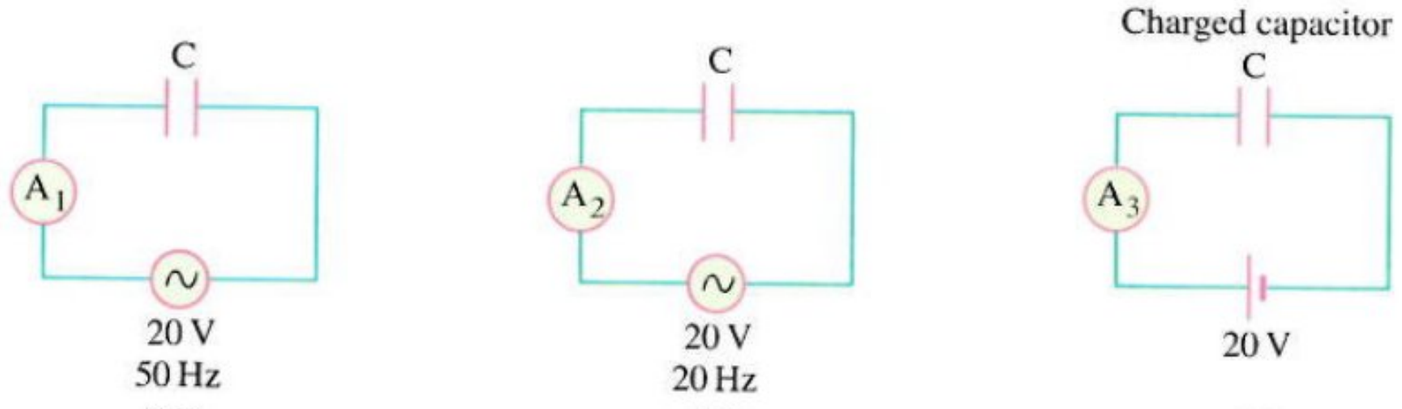
(c) increasing the speed of rotation of the dynamo's coil?

(d) inserting an iron core inside the coil?

(e) replacing the dynamo's coil with another one that has double the number of turns of the first?

C circuit

12 The following figure represents three electric circuits (1), (2) and (3), each of them has a capacitor of capacitance (C), an electric source of voltage 20 V and a hot-wire ammeter,



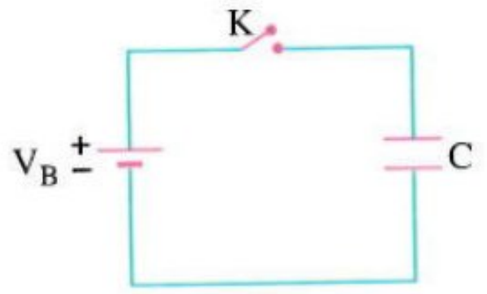
explain each of the following:

(a) The reading of ammeter A_3 equals zero while the readings of both ammeters A_1, A_2 doesn't equal zero.

(b) The reading of ammeter A_1 is greater than the reading of ammeter A_2

Scanned with OKEN Scanner

13 In the opposite electric circuit, a capacitor (C), a battery V_B of negligible internal resistance and a switch K are connected together in series, when the switch K is closed, the capacitor C takes time (t) to complete the charging process. **Describe what happens within the interval t for each of:**

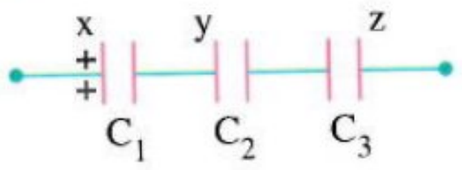


(a) The flow rate of the electric charge from the battery to the capacitor.

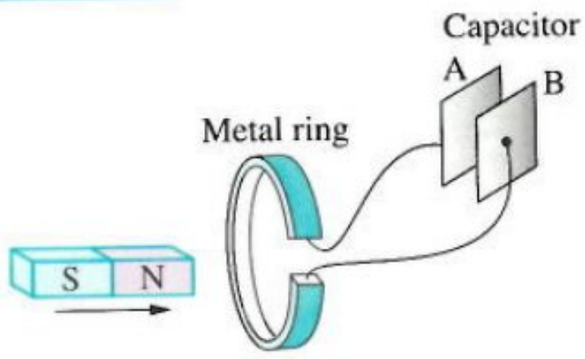
(b) The potential difference between the plates of the capacitor.

(c) The potential difference between the poles of the battery.

14 Three capacitors of capacitances C_1, C_2, C_3 are connected in series. The plate x of capacitor C_1 is charged with a positive charge of quantity Q as in the opposite figure. **Determine** the type and the quantity of the charge which is formed on the plates y, z.

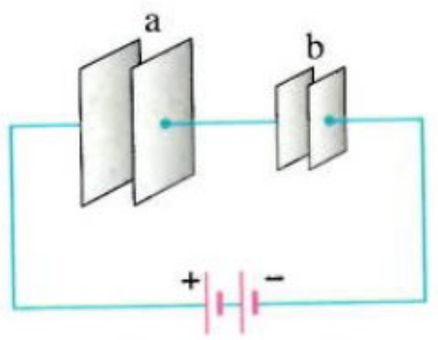


15 **What's** the type of the electric charge that is formed on plate A of the capacitor when moving the magnet towards the metal ring that is shown in the opposite figure? **And why**?

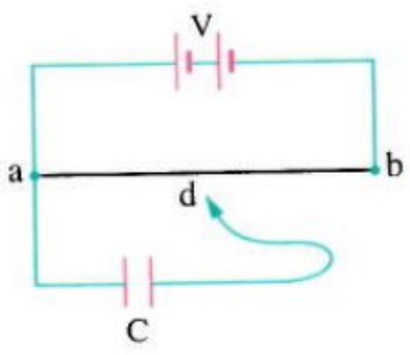


16 Three capacitors of different capacitances C_1, C_2, C_3 are connected in parallel with a DC source of potential difference V, **prove that** after the complete charging of capacitors, the accumulated quantity of charges on each of the capacitors differs from the others.

17 Two capacitors a, b of different capacitances due to their different plates areas are connected in series with a battery, as in the opposite figure. **Does** the quantity of charge that is accumulated on the plates of the two capacitors differ? **And why**?



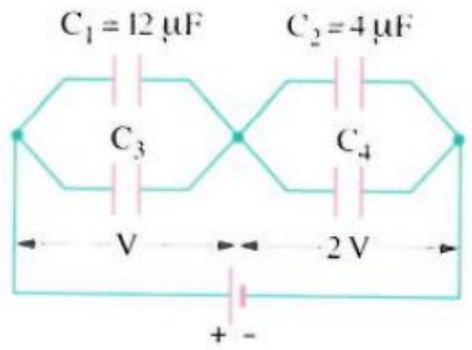
18 The opposite electric circuit is used to determine the capacitance of a capacitor (C) where the applied potential difference between the plates of the capacitor changes by moving the slider d on the wire ab. **Represent graphically** the relation between the accumulated charge (Q) on the capacitor on the vertical axis and the potential difference (V) between its plates on the horizontal axis. **How** can the capacitance of the capacitor be determined from the graph representing this relation?



CamScanner
Scanned with OKEN Scanner

CamScanner

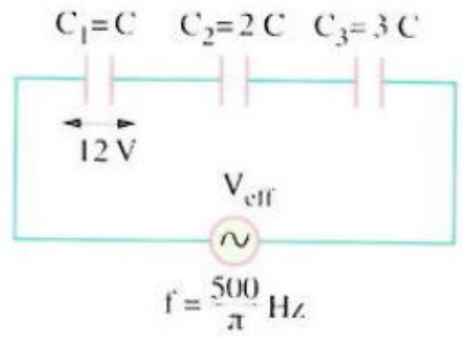
19 A group of four capacitors C_1, C_2, C_3, C_4 are connected to DC source, if the charge on capacitor C_1 equals the charge on capacitor C_3, **find** the capacitance of capacitor C_4



20 In the opposite electric circuit, if the effective value of AC current through the circuit is 1 A, **calculate:**

(a) The value of capacitance C

(b) The effective emf of the source (V_{eff}) with neglecting the internal resistance.

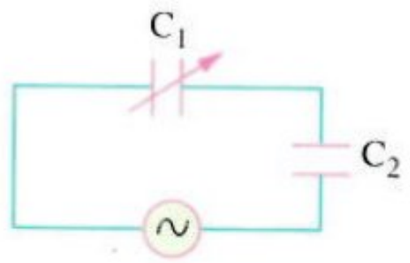


21 In the opposite electric circuit, when increasing the capacitance of capacitor C_1, **what will happen to each of:**

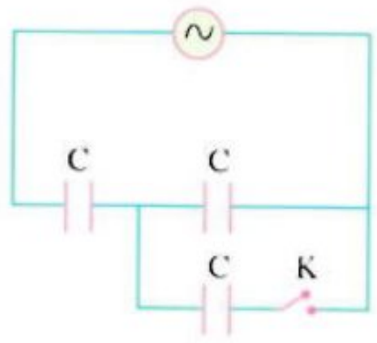
(a) the phase angle between the total voltage and the current in the circuit?

(b) the value of the current through the source?

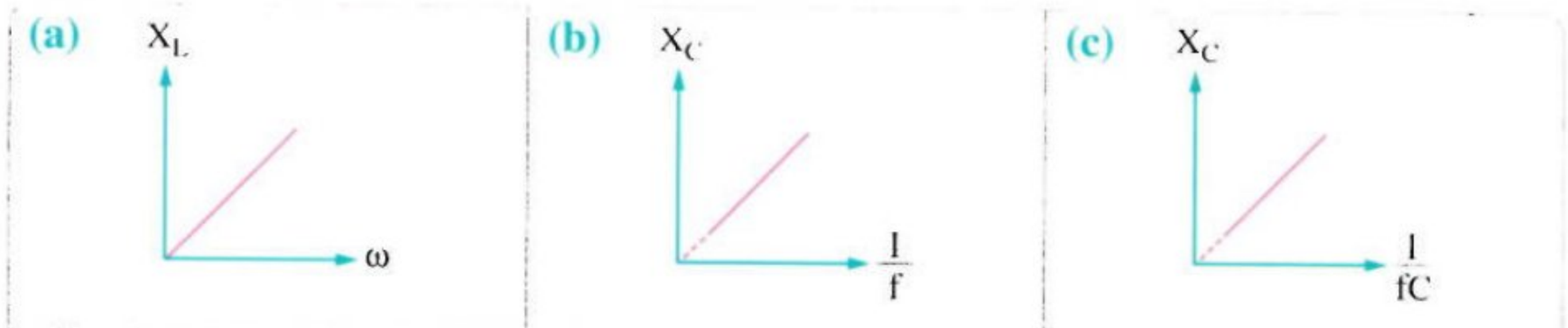
(c) the voltage across the plates of capacitor C_2?



22 In the opposite electric circuit with neglecting the ohmic resistance in the circuit, **calculate** the ratio between the two values of currents in the circuit before and after closing the switch $\left(\frac{I_1}{I_2}\right)$.



23 **Write the mathematical relations and the equivalent of the slope for each of the following graphs:**



Where: (X_L) the inductive reactance of a coil, (ω) the angular speed, (X_C) the capacitive reactance of a capacitor, (f) frequency, (C) the capacitance of capacitor.

Scanned with OKEN Scanner

Questions of the Previous Years Exams ? on Lesson 1

Hot-wire ammeter

1 In the hot wire ammeter, the wire is mounted on a metallic plate of the same expansion coefficient as the wire to ..... . *(Exp. May 21)*

(a) make the pointer return back quickly to zero position as current is turned off

(b) decrease the efficiency of the device

(c) avoid the zero error

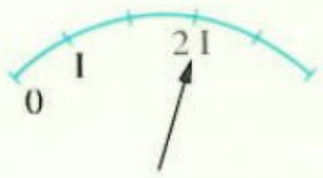
(d) increase the thermal expansion of the wire

2 During the calibration of a hot wire ammeter scale, when an AC current of effective value I passes through the device, its pointer gets deflected to the position shown in the opposite figure, so which one of the following figures represents the correct position of the pointer of the hot wire ammeter when the effective value of the passing current becomes 2I? *(Exp. June 21)*

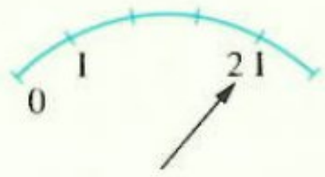




(a)



(b)

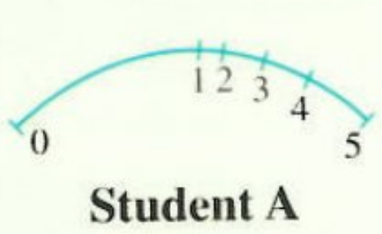


(c)

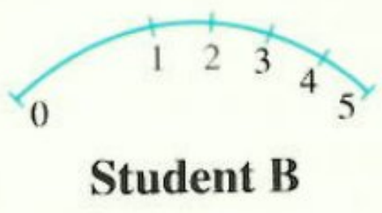


(d)

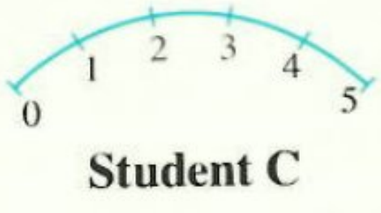
3 Four students drew schematic diagrams for the hot wire ammeter scale, which student drawing is the correct scale for the hot wire ammeter? *(1st session 21)*



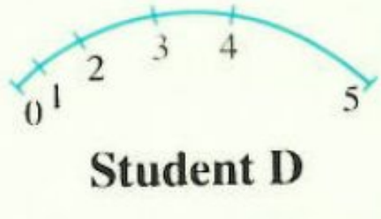
Student A



Student B



Student C



Student D

(a) Student A (b) Student B (c) Student C (d) Student D

4 In a hot wire ammeter, the heat energy generated in the platinum-iridium wire as the alternating electric current flows is directly proportional to the ..... . *(2nd session 21)*

(a) $\frac{1}{V_{eff}^2}$ (b) I_{eff} (c) I_{max} (d) V_{eff}^2

CamScanner

Scanned with OKEN Scanner

CamScanner

5 It is noticed that, the pointer of a hot wire ammeter moves on a non-uniform scale. This is due to ..... .

(1st session 22)

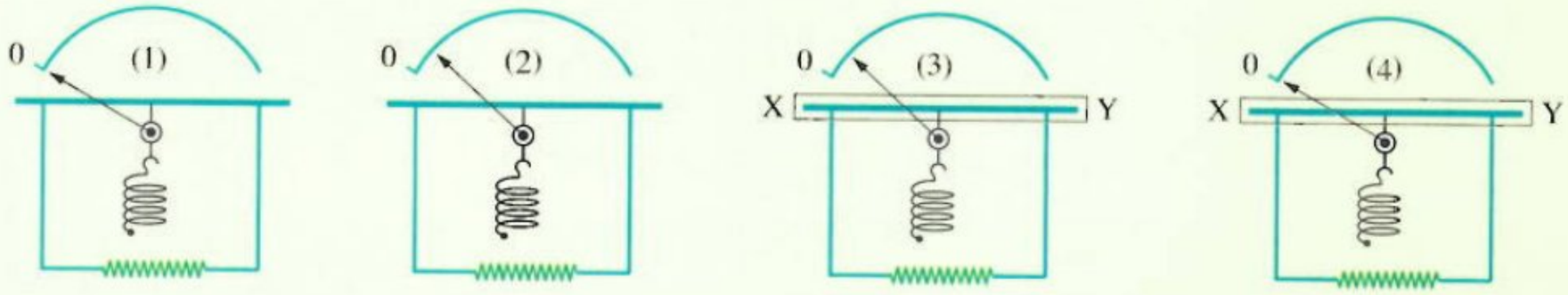
(a) the hot wire ammeter measures the maximum value of the alternating electric current

(b) the pointer of the hot wire ammeter moves slowly at the beginning of passage of the electric current

(c) the amount of heat energy generated is directly proportional to the intensity of the electric current

(d) the amount of heat energy generated is directly proportional to the square of the intensity of the electric current

6 In one of the hot climate countries, a student use the hot wire ammeter in the school lab which is not airconditioned.



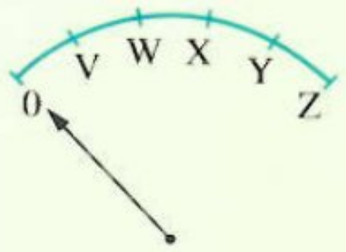
Which two figures show the correct position of the hot wire ammeter pointer at the lab's temperature?

(Knowing that (XY) is a plate made of a material having the same expansion coefficient of platinum and iridium wire)

(2nd session 22)

(a) (2), (4) (b) (1), (3) (c) (3), (2) (d) (4), (1)

7 The figure illustrates the scale of a hot wire ammeter where the spaces between the positions marked on the scale are equal. When a current of intensity (I) passes through the wire of the device, the pointer deflects to the position V. Which of the following choices represent the current intensity passing through the wire of the device when the pointer deflects to the position Y?



(Exp. 23)

(a) 2 I (b) 3 I (c) 4 I (d) 5 I

8 If the shunt resistor in the hot wire ammeter is replaced by another of smaller value while keeping the effective value of the electric current flowing through the circuit constant, then ..... .

(1st session 23)

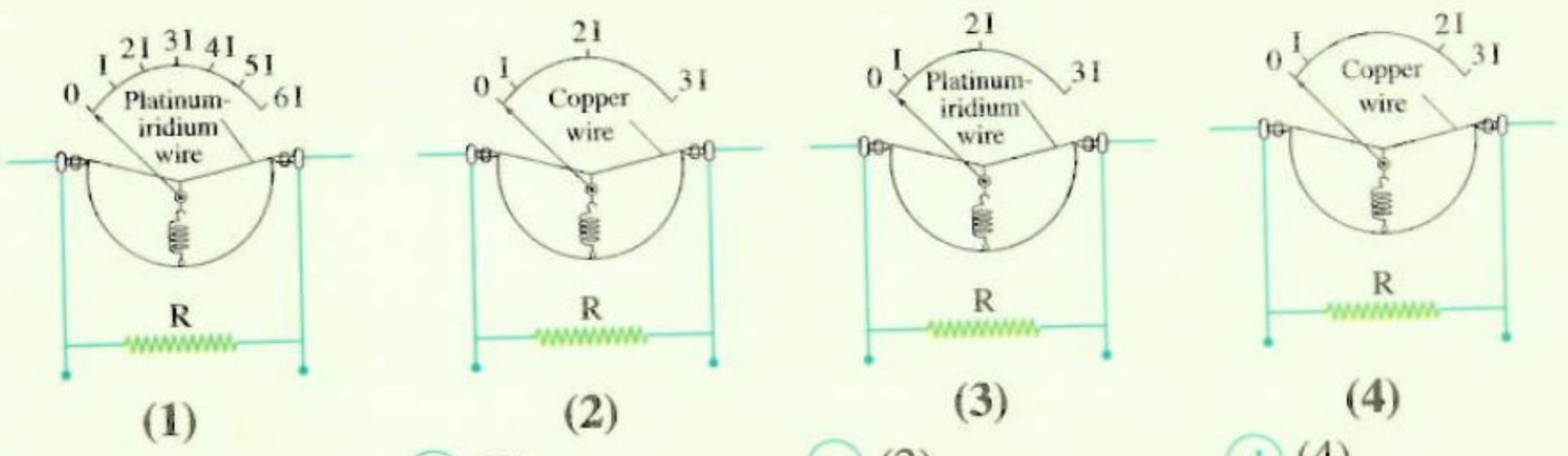
	Thermal power generated in the platinum-iridium wire	Total resistance of the ammeter
(a)	decreases	increases
(b)	decreases	decreases
(c)	increases	decreases
(d)	increases	increases

Scanned with OKEN Scanner

9 An electric current of intensity (I) flows through a hot-wire ammeter. If the current through the hot-wire ammeter increases to (2I), ... *(2nd session 23)*

	The expansion of iridium-platinum wire	The generated heat energy in unit time
a	increases	increases to double
b	decreases	decreases to its half
c	increases	increases to four times
d	decreases	decreases to its quarter

10 Which of the following figures represents correctly the structure of the hot wire ammeter? *(1st session 24)*

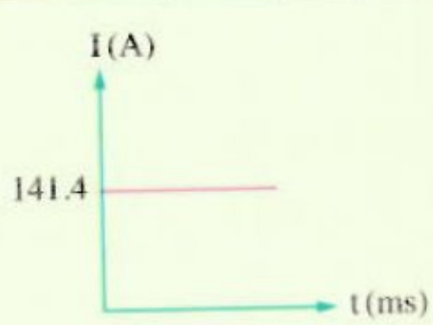


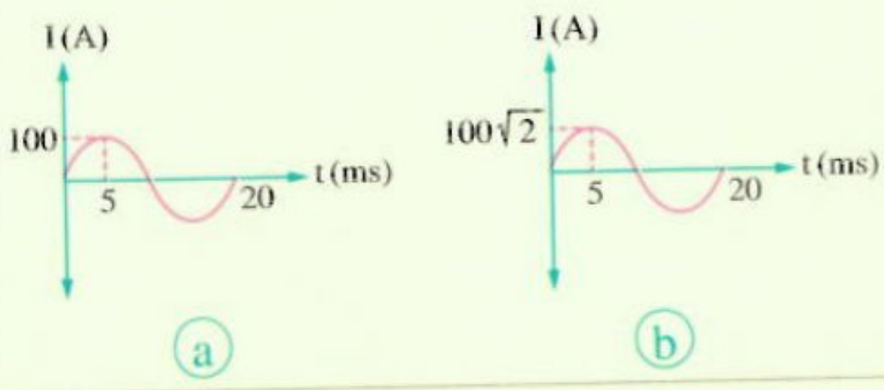
a (1) b (3) c (2) d (4)

11 If the shunt resistor in the hot wire ammeter is replaced by another of smaller value while keeping the effective value of the electric current flowing through the circuit constant, then: *(2nd session 24)*

	Thermal power generated in the platinum-iridium wire	Total resistance of the ammeter
a	increases	increases
b	decreases	decreases
c	increases	decreases
d	decreases	increases

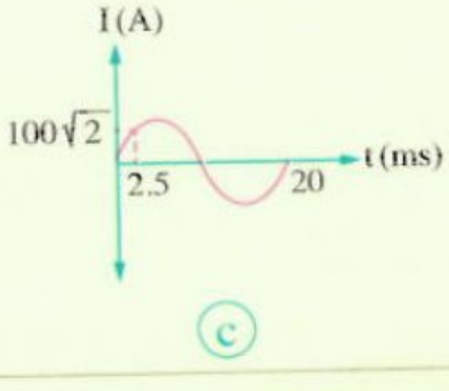
12 The figure shows the relation between the intensity of direct current and the time, which figure shows the value of AC which produces the same thermal energy in the same resistance through the same time as the direct current? *(2nd session 24)*



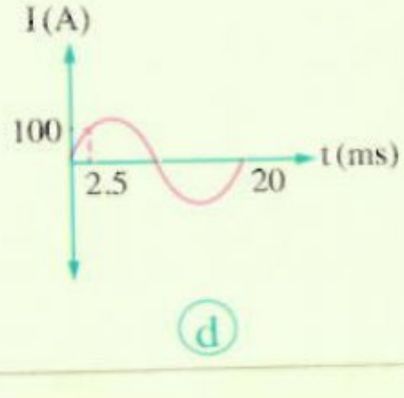


a

b



c



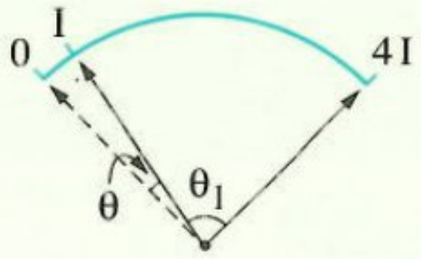
d

CamScanner

Scanned with OKEN Scanner

CamScanner

13 The opposite figure represents two deflections of a hot-wire ammeter pointer from the zero position, so the value of θ_1 in terms of θ equals ..... .



(Practice 25)

(a) 5 θ (b) 10 θ (c) 15 θ (d) 20 θ

14 The alternating current **cannot** be used directly in ..... . *(Practice 25)*

(a) lighting lamps (b) operating household appliances
(c) charging batteries (d) operating electrical transformers

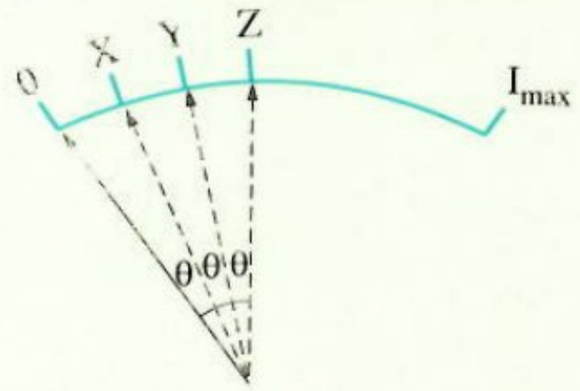
15 When a current of effective alternating current (I) passed through a platinum-iridium wire, the consumed electrical power in the platinum-iridium wire was (P_w). So, the effective value of the alternating current that makes the consumed thermal power in the same wire ($2 P_w$) equals ..... . *(Practice 25)*

(a) I (b) $\sqrt{2}$ I (c) 0.5 I (d) 2 I

16 In the hot-wire ammeter, when replacing the platinum-iridium wire with another wire made of a metal with a lower thermal expansion coefficient and setting up the same intensity of electric current through it, the deviation angle of the indicator ..... . *(Practice 25)*

(a) increases (b) decreases (c) does not change (d) vanishes

17 The opposite figure shows the ammeter scale and several current values, where:

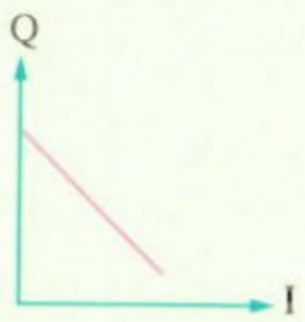


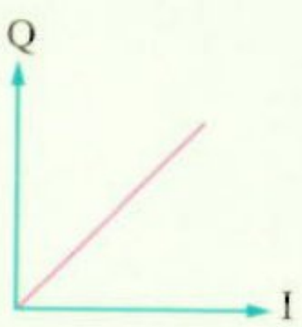
- (ΔI_1) represents the change in current value from zero to position X.
- (ΔI_2) represents the change in current value from position X to position Y.
- (ΔI_3) represents the change in current value from position Y to position Z.

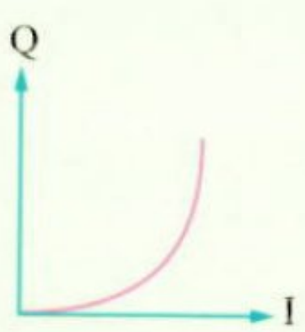
So, if the shown angles are equal, then the correct arrangement for the changes in current value is ..... *(Practice 25)*

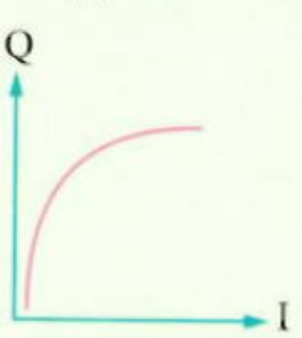
(a) $\Delta I_1 > \Delta I_2 > \Delta I_3$ (b) $\Delta I_1 = \Delta I_2 = \Delta I_3$ (c) $\Delta I_1 < \Delta I_2 < \Delta I_3$ (d) $\Delta I_1 < \Delta I_2 > \Delta I_3$

18 Which of the following graphs illustrates the relationship between the amount of heat (Q) generated in a platinum-iridium wire and the effective value of the current (I)? *(Practice 25)*

(a)


(b)


(c)


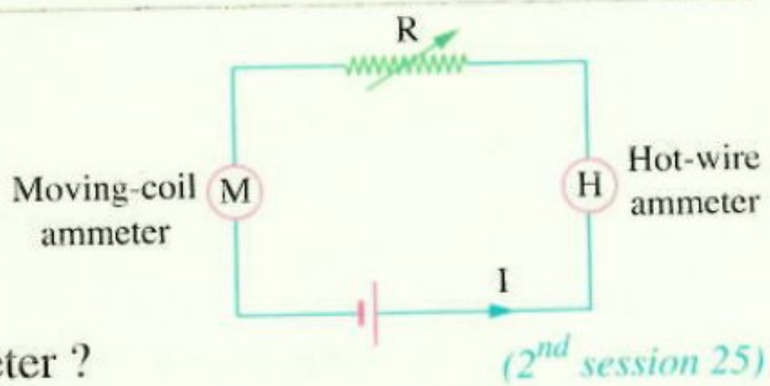
(d)


Scanned with OKEN Scanner

19 The reading of a hot-wire ammeter in an AC circuit represents ..... . *(1st session 25)*

(a) the average value of a complete cycle of AC current
(b) the maximum value of AC current
(c) the instantaneous value of AC current
(d) the effective value of AC current

20 The figure shows an electric circuit consisting of a DC power supply, a variable resistor (R) a hot-wire ammeter (H), and a moving coil ammeter (M). If the current intensity changes with equal amounts, what happens to the pointer of each ammeter ?



(2nd session 25)

	In hot-wire ammeter (H)	In moving coil ammeter (M)
(a)	Deflects uniformly	Deflects uniformly
(b)	Deflects uniformly	Deflects non-uniformly
(c)	Deflects non-uniformly	Deflects uniformly
(d)	Deflects non-uniformly	Deflects non-uniformly

21 You have four different electric circuits, each contains a hot wire ammeter where each of the following graphs w, x, y, z represents the relation between the intensity of the current (I) which is passing in the ammeter and time (t),

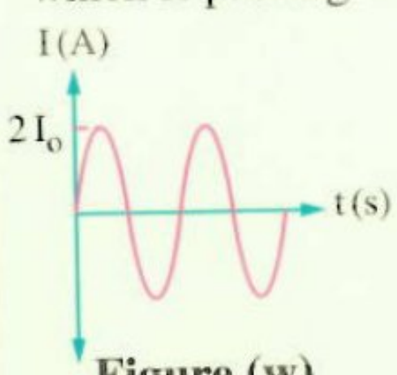


Figure (w)

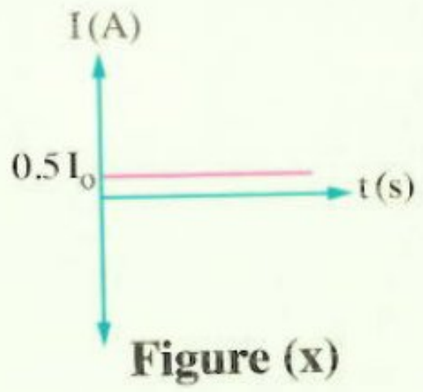


Figure (x)

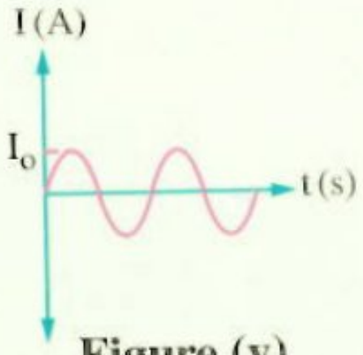


Figure (y)

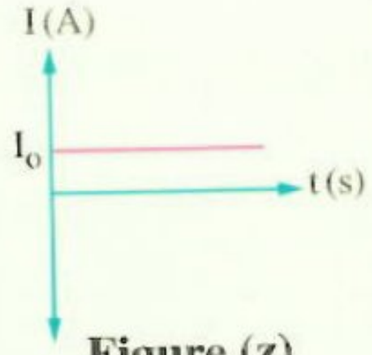


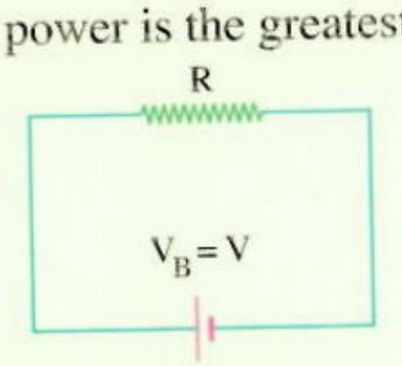
Figure (z)

so, the correct arrangement for these graphs according to the hot-wire ammeter reading is ..... *(2nd session 25)*

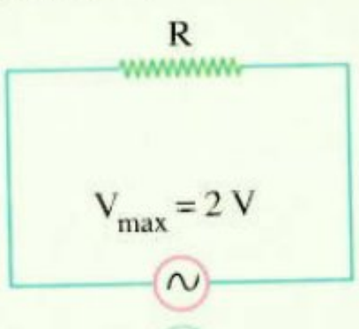
(a) $z < x < y < w$
(b) $w < z < y < x$
(c) $x < y < z < w$
(d) $x < y = z < w$

R circuit

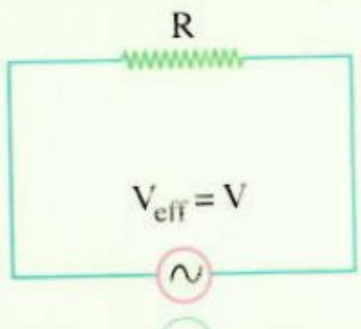
22 The following figures represent four electrical circuits, each contains a resistance R and an electrical source of negligible ohmic resistance, then the circuit in which the consumed power is the greatest is ..... . *(Practice 25)*



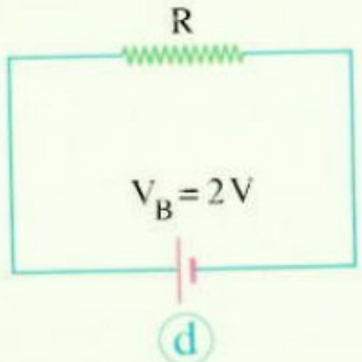
(a)



(b)



(c)



(d)

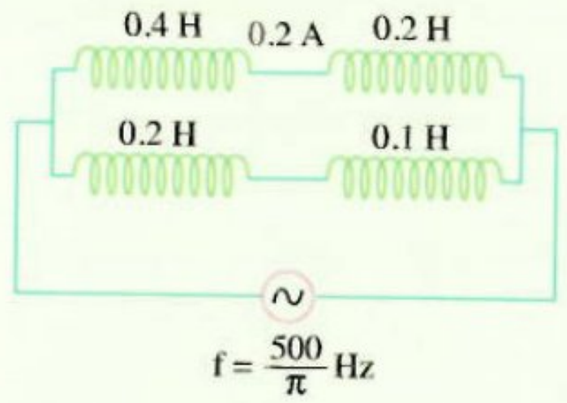
CamScanner
Scanned with OKEN Scanner

CamScanner

L circuit

23 From the given data in the figure, the value of the electromotive force of the source of alternating current equals ..... .

(1st session 23)

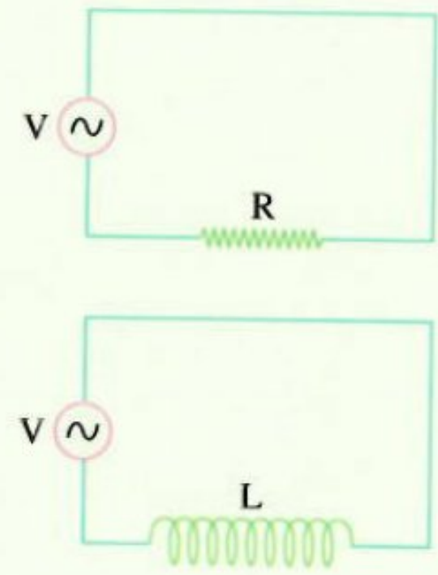


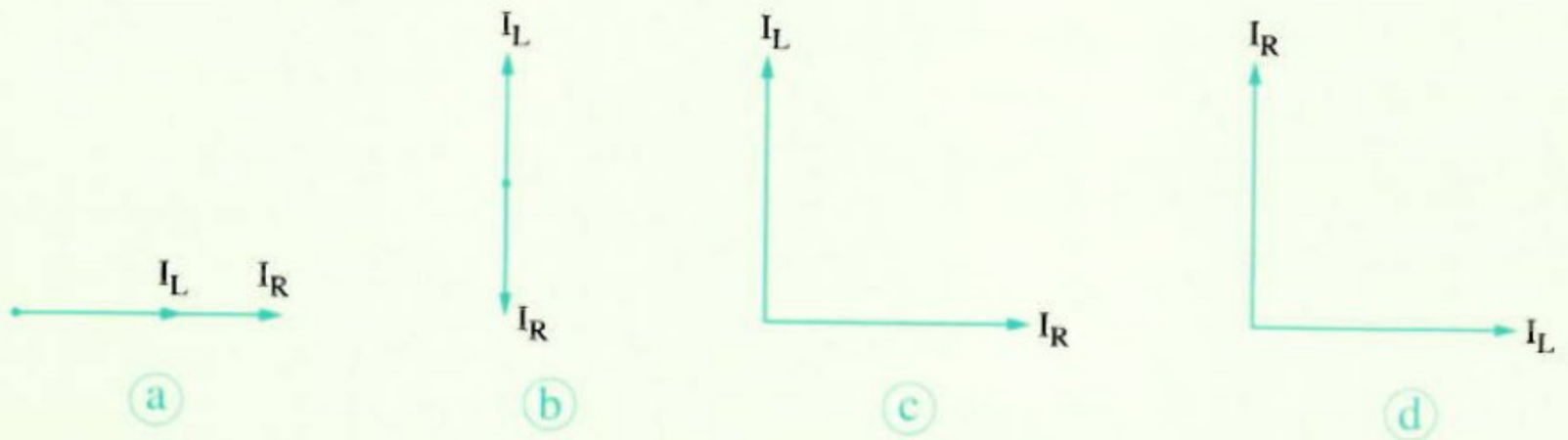
(a) 20 V (b) 40 V

(c) 120 V (d) 80 V

24 The figure illustrates two AC circuits, one contains an ohmic resistance (R) while the other contains an inductive coil (L) of negligible ohmic resistance. Assuming that the voltages of the two sources are in the same phase, so the diagram below that represents the phase difference between the currents I_R and I_L is ..... .

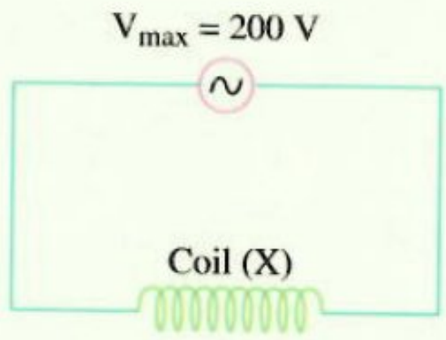
(Exp. May 21)





25 The opposite figure shows an AC source, whose maximum voltage is 200 V and its frequency is 50 Hz, connected to an inductive coil (X) of self-inductance (L) and negligible ohmic resistance. Given that the effective value of current passing in the circuit is 2 A, what are the value of self-inductance and the method of connecting another coil with coil (X) to double the effective value of the current in the circuit?

(Exp. May 21, Practice 25)



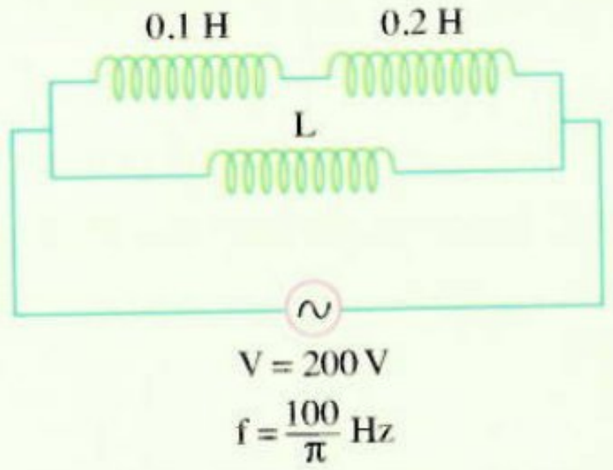
(a) 0.22 H, in series. (b) 0.22 H, in parallel.

(c) 0.32 H, in series. (d) 0.32 H, in parallel.

Scanned with OKEN Scanner

26 Three inductors of negligible ohmic resistances are connected together as in the figure, given that the effective value of the passing current in the circuit is 5 A and with neglecting the mutual induction between the coils, then the value of inductance L equals ..... .

(Exp. June 21)



(a) 0.6 H (b) 0.4 H (c) 0.3 H (d) 1 H

27 Several identical inductors of negligible ohmic resistance were connected together in series to an AC supply of frequency $\frac{50}{\pi}$ Hz, their total inductive reactance became 40 Ω, but when they are connected together in parallel to the same AC source, the total inductive reactance becomes 2.5 Ω, neglecting the mutual induction between the coils, so the self-inductance for each coil = ...

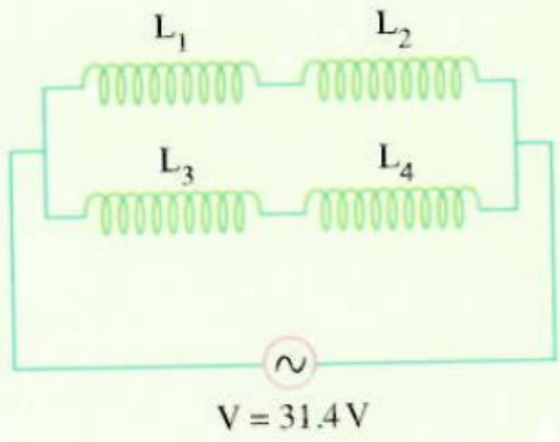
(1st session 21)

(a) 0.1 H (b) 0.2 H (c) 0.3 H (d) 0.4 H

28 Four inductive coils of negligible ohmic resistance each of them has a self-inductance of 50 mH were connected together as in figure. If the effective value of the passing current in the circuit is 10 A, so the frequency of this current = ...

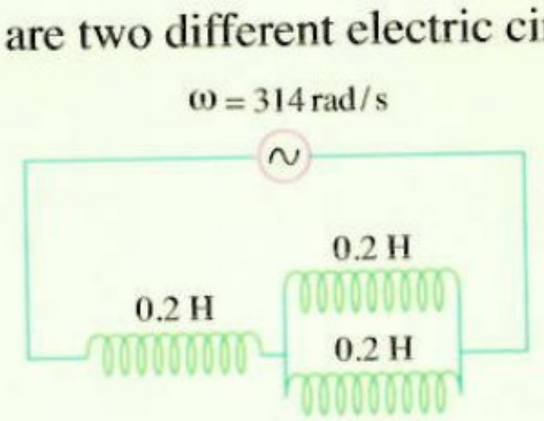
(2nd session 21)

(Neglecting the mutual induction between the coils)

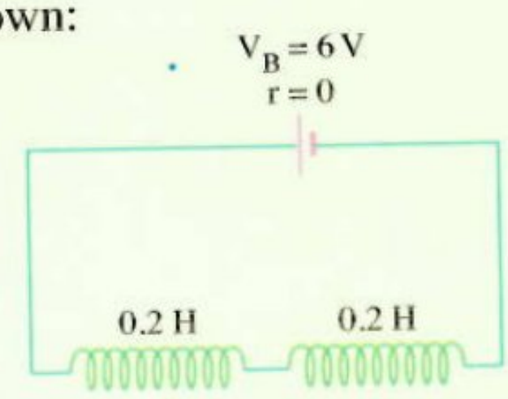


(a) 20 Hz (b) 50 Hz (c) 10 Hz (d) 60 Hz

29 (A) and (B) are two different electric circuits as shown:



Circuit (A)



Circuit (B)

The inductive reactance for circuit A = ...

While that for circuit B = ...

(a) 94.2 Ω, zero Ω (b) 94.2 Ω, 125.6 Ω

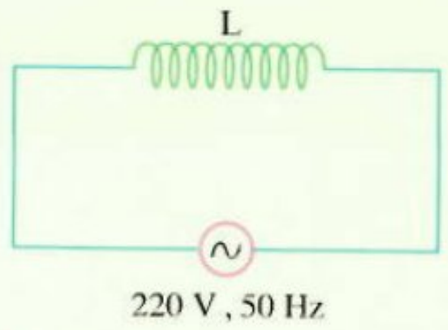
(c) 62.8 Ω, zero Ω (d) 62.8 Ω, 125.6 Ω *(1st session 22)*

CamScanner

Scanned with OKEN Scanner

CamScanner

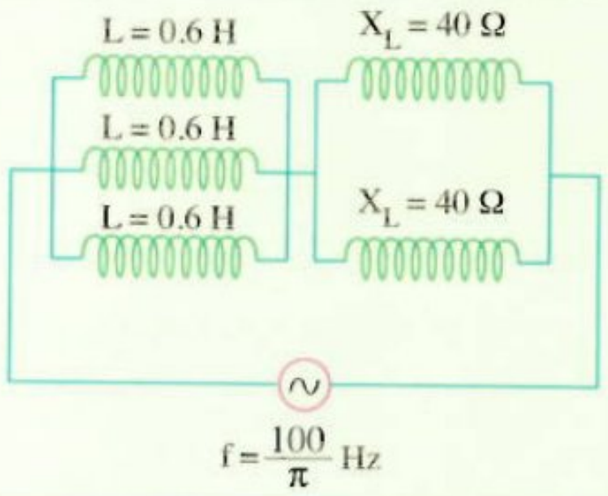
30 In the shown circuit, an alternating current source (220 V, 50 Hz) is connected to an inductive coil of self-inductance (L) with negligible ohmic resistance. Current of intensity 2 A passes in the coil. So, the self-inductance of the coil (L) equals ..... . ($\pi = 3.14$) *(1st session 22)*



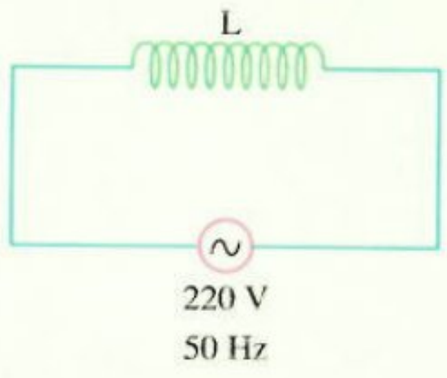
(a) 0.7 H (b) 0.35 H (c) 4.4 H (d) 0.04 H

31 In the shown electric circuit, the equivalent inductive reactance equals to ..... . *(2nd session 22)*

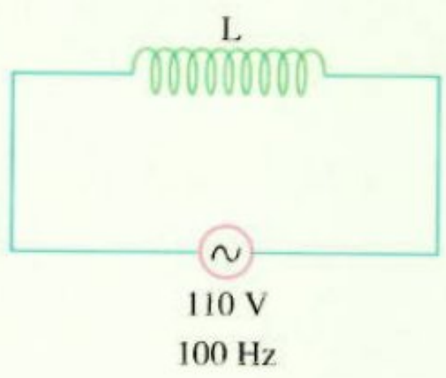
(a) 40 Ω
(b) 60 Ω
(c) 20 Ω
(d) 80 Ω



32 In the figure, an inductive coil of self-inductance (L) and negligible ohmic resistance is inserted in two alternating current circuits (1), (2).



Circuit (1)

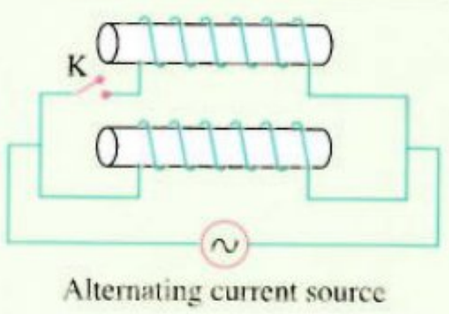


Circuit (2)

The ratio between : $\frac{\text{the electric current in circuit (1)}}{\text{the electric current in circuit (2)}}$ = ... *(2nd session 22)*

(a) $\frac{1}{1}$ (b) $\frac{2}{1}$ (c) $\frac{4}{1}$ (d) $\frac{1}{2}$

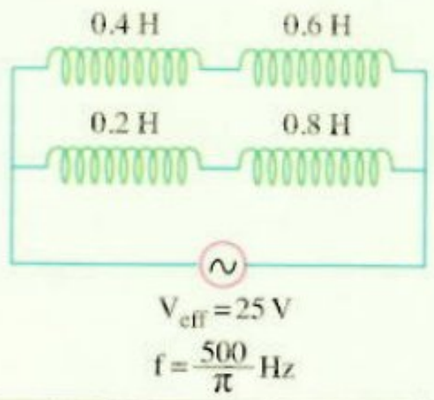
33 The circuit shows two inductors of negligible ohmic resistance connected to an alternating current source. When the key (K) is closed the phase angle between the voltage and the current is ..... . *(Exp. 23)*



(a) 180° (b) 90° (c) 45° (d) zero

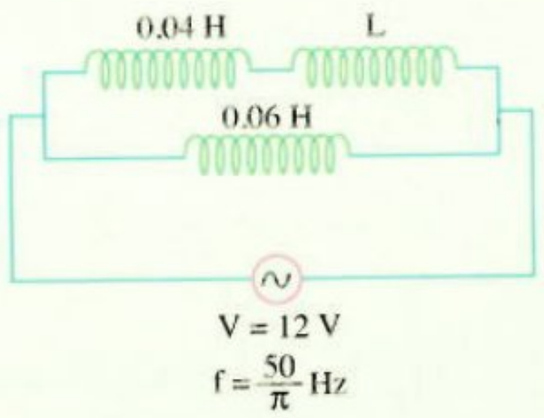
34 According to the shown electric circuit, the effective value of the alternating current in the circuit is ..... . *(Exp. 23)*

(a) 0.05 mA (b) 0.5 mA
(c) 5 mA (d) 50 mA



Scanned with OKEN Scanner

35 Three coils of negligible ohmic resistance are connected as in the figure with a source of alternating current.



So, the value of the unknown self induction coefficient (L) which allows an electric current of 3 A to pass in the circuit is ..... .

(Neglect the mutual induction between the coils) *(2nd session 23)*

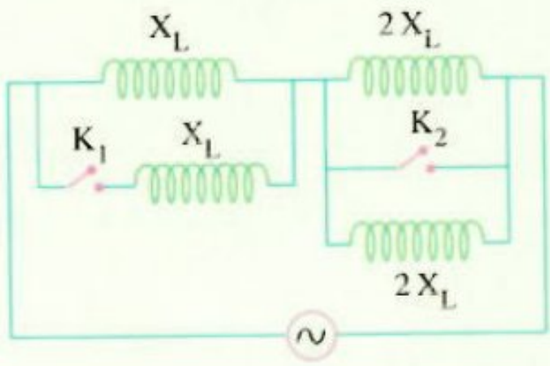
(a) 0.08 mH (b) 80 mH (c) 40 mH (d) 120 mH

36 From the shown figure, the ratio between:

$$\frac{\text{The total inductive reactance when } K_1 \text{ is closed while } K_2 \text{ is open}}{\text{The total inductive reactance when } K_2 \text{ is closed while } K_1 \text{ is open}}$$

= .....

(1st session 24)



(a) $\frac{1}{3}$ (b) $\frac{2}{3}$

(c) $\frac{3}{1}$ (d) $\frac{3}{2}$

37 From the opposite figures, by neglecting the ohmic resistance for the coils and the mutual induction between the coils.

So, the ratio between $\left(\frac{I_2}{I_1}\right)$ = ...

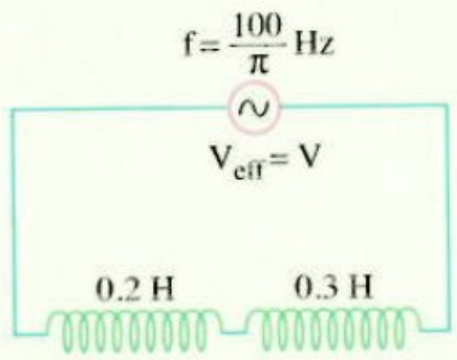


Figure (1)

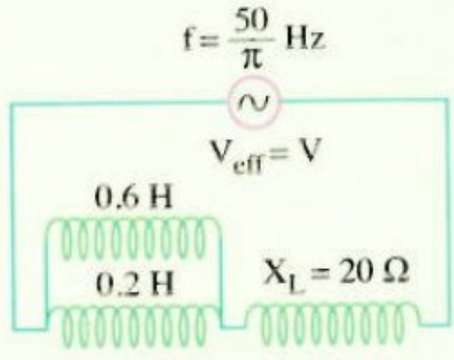


Figure (2)

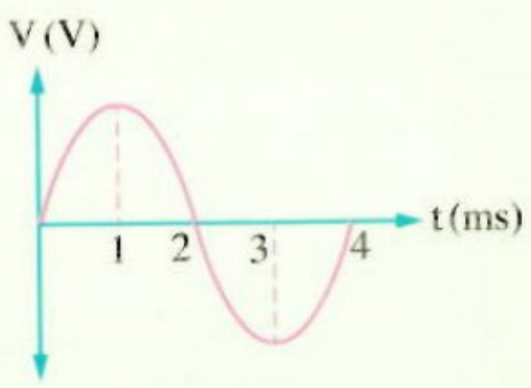
(2nd session 24)

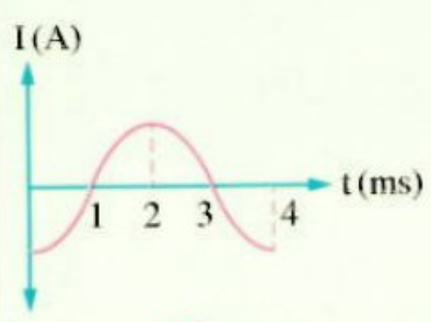
(a) $\frac{20}{7}$ (b) $\frac{7}{20}$

(c) $\frac{20}{3}$ (d) $\frac{3}{20}$

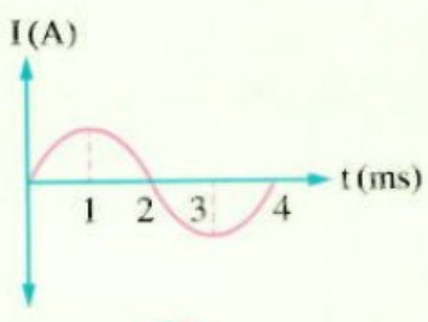
38 If the potential difference (V) across an inductor of no ohmic resistance connected to an AC source varies with time (t) as shown in the opposite graph, then the graph that represents the variation of the current (I) through the inductor with time (t) during the same period is ..... .

(Practice 25)

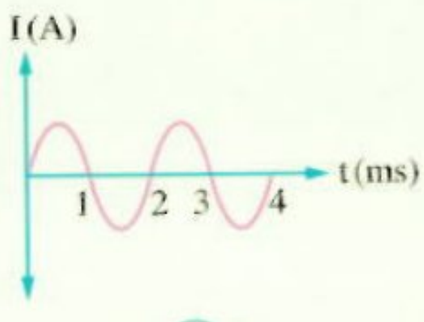




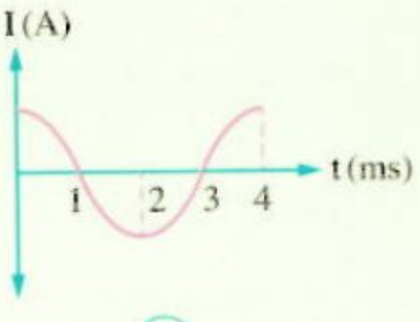
(a)



(b)



(c)



(d)

CamScanner

Scanned with OKEN Scanner

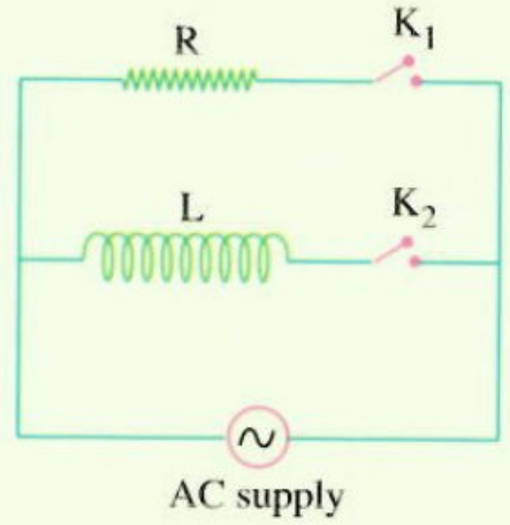
CamScanner

39 If the inductive reactance of a coil = 440 L ohms, where (L) is the self-inductance of the coil, then the angular velocity is ..... radians/second. *(Practice 25)*

(a) 440 (b) 140 (c) 70 (d) 80

40 The figure shows an AC circuit consisting of an AC source, an ohmic resistor (R), and an inductor (L) of negligible ohmic resistance.

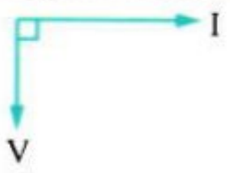
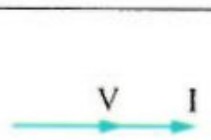
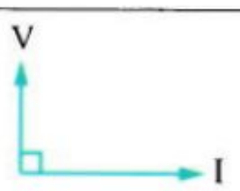
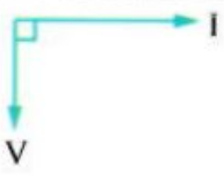
What is the phase angle between the voltage across the source and the current in the circuit in the following two cases. *(1st session 25)*



	When switch (K_1) is only closed	When switch (K_2) is only closed
(a)	90°	90°
(b)	0°	90°
(c)	90°	0°
(d)	0°	0°

41 Two AC circuits; circuit (1) consists of an AC source and an ohmic resistance, and circuit (2) consists of an AC source and an inductor of negligible ohmic resistance.

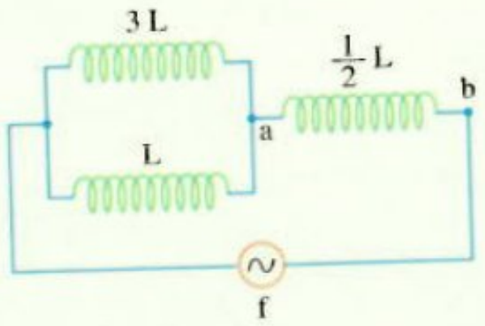
Which of the following choices represents the phase angle between the total voltage and current in each circuit? *(2nd session 25)*

	Circuit (1)	Circuit (2)
(a)		
(b)		
(c)		
(d)		

Scanned with OKEN Scanner

42 The figure represents an alternating current source of frequency f connected to several inductors of negligible ohmic resistance. If the electric current passing in the inductor 3L is I, the potential difference between points a and b equals ..... .

(Practice 26)



(a) 4 IπLf (b) 2 IπLf (c) $\frac{1}{2}$IπLf (d) $\frac{1}{4}$IπLf

C circuit

43 A 10 μF capacitor is connected to a 1000 Hz oscillator of maximum emf 5 V, hence the maximum intensity of the current that passes in the capacitor circuit is approximately equal to ..... .

(Exp. May 21)

(a) 0.6 A (b) 1.2 A (c) 0.8 A (d) 0.3 A

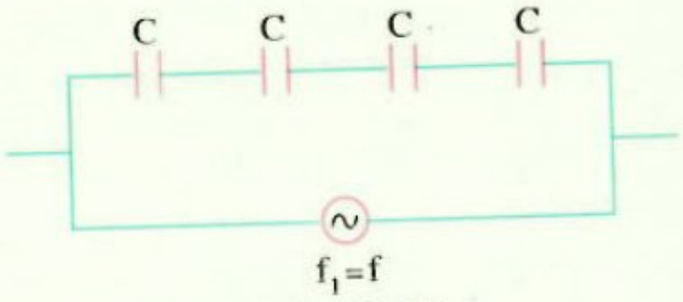
44 In the two electric circuits, if the capacitance of each capacitor is C, the ratio of

$$\frac{\text{the equivalent capacitive reactance in circuit (1)}}{\text{the equivalent capacitive reactance in circuit (2)}}$$

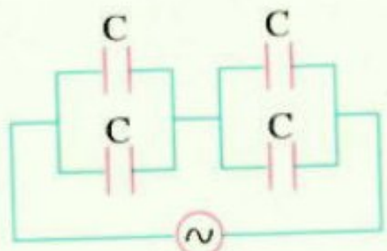
= .....

(Exp. June 21)

(a) $\frac{1}{2}$
(b) $\frac{2}{1}$
(c) $\frac{8}{1}$
(d) $\frac{1}{8}$



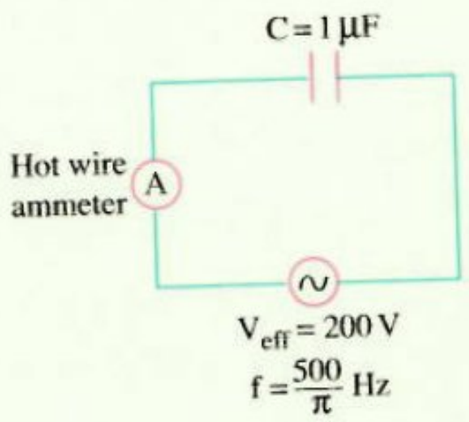
Circuit (1)



Circuit (2)

45 From the opposite figure which represents an electric circuit that contains an AC source, a hot wire ammeter of negligible resistance and a capacitor, the reading of the hot wire ammeter equals ..... .

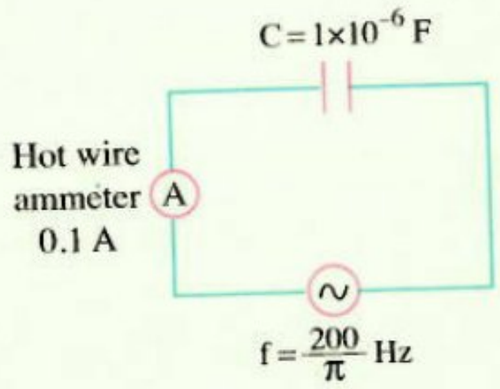
(Exp. June 21)



(a) 0.02 A (b) 0.2 A
(c) 2 A (d) 20 A

46 The opposite figure represents an electric circuit that contains a hot wire ammeter of negligible ohmic resistance, a capacitor and an AC source, so the effective value of the emf of the source equals ..... .

(1st session 21)



(a) 250 V (b) 2.5 V
(c) 25 V (d) 2500 V

CamScanner

Scanned with OKEN Scanner

CamScanner

47 The following four figures illustrate four similar capacitors, each of capacitance C.

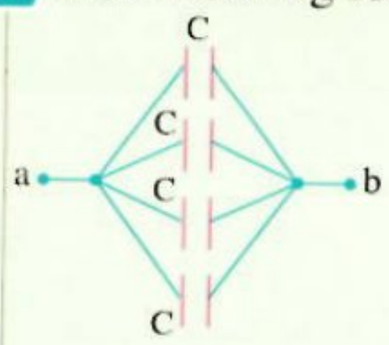


Figure (1)

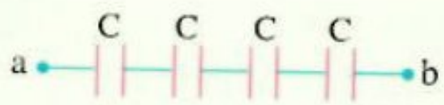


Figure (2)

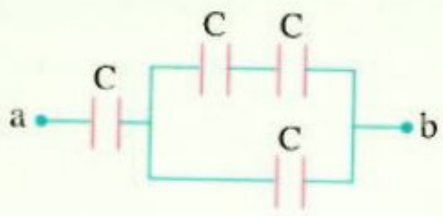


Figure (3)

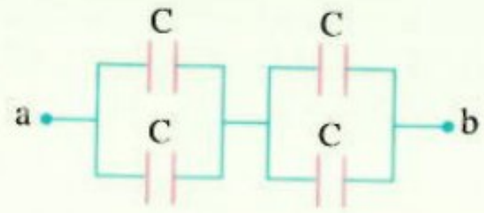


Figure (4)

So, which figure should be connected between the points (a, b) to close the opposite electric circuit so that the electric current that passes through the circuit has the maximum value? *(1st session 21)*



(a) Figure (1) (b) Figure (2) (c) Figure (3) (d) Figure (4)

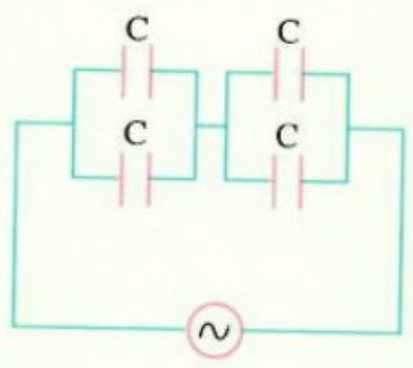
48 In the two electric circuits, if the capacitance of each capacitor is C, so the ratio:

$$\frac{\text{the capacitive reactance in circuit (2)}}{\text{the capacitive reactance in circuit (1)}} = \ldots\ldots\ldots$$

(2nd session 21)

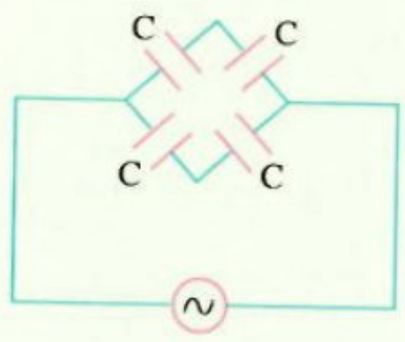
(a) $\frac{2}{1}$ (b) $\frac{1}{4}$

(c) $\frac{4}{1}$ (d) $\frac{1}{2}$



$f_1 = f$

Circuit (1)



$f_2 = 2f$

Circuit (2)

49 The four figures show three similar capacitors, each of capacitance (C). *(1st session 22)*

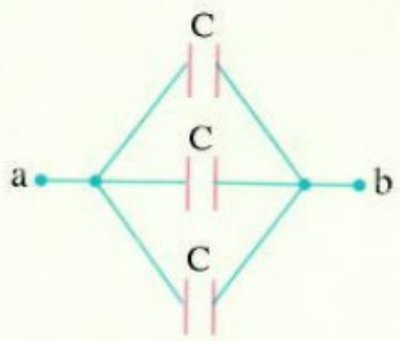


Figure (1)

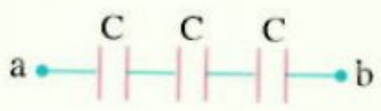


Figure (2)

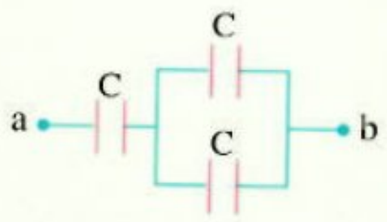


Figure (3)

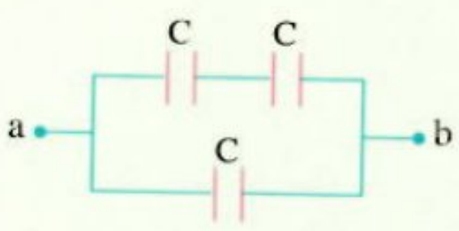
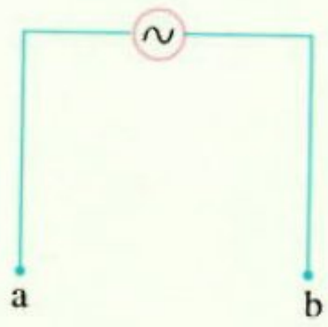


Figure (4)

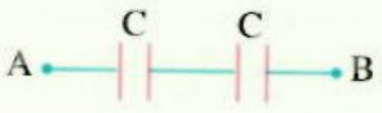
Which figure should be connected between points (a, b) to close the electric circuit shown making the electric current to pass through it with the least value?

(a) Figure (1) (b) Figure (2)

(c) Figure (3) (d) Figure (4)



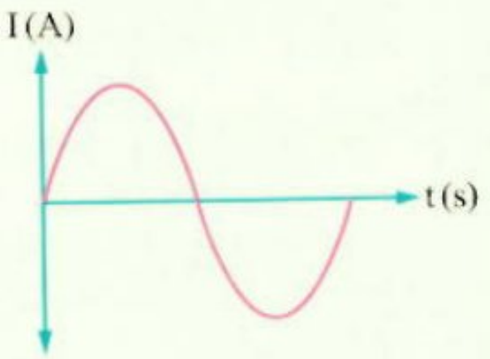
50 The figure illustrates two capacitors of the same capacitance (C) connected in series. When a third capacitor of capacitance that equals half the capacitance of one of the two capacitors is connected in parallel between (A, B), the total capacitance of the three capacitors equals ..... . *(2nd session 22)*

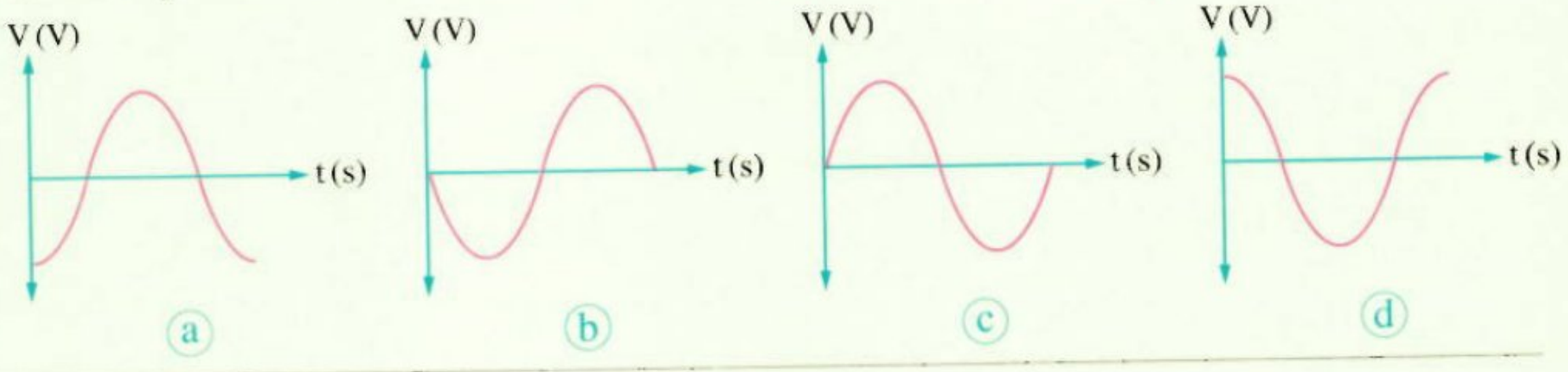


(a) C (b) 2 C (c) $\frac{C}{2}$ (d) $\frac{3}{2}$ C

Scanned with OKEN Scanner

51 The graph illustrates the relation between the change in the current intensity (I) and the time (t) in an alternating current circuit contains a capacitor. Which of the following graphs represents the change of voltage across the two plates of the capacitor at same time? *(Exp. 23)*

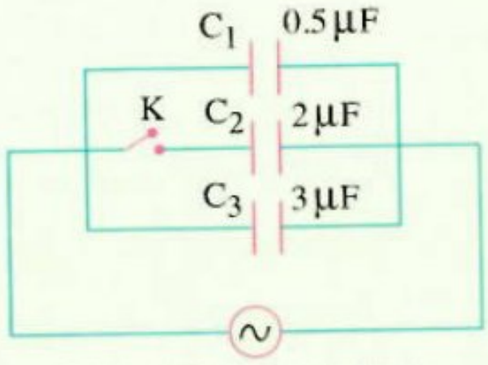




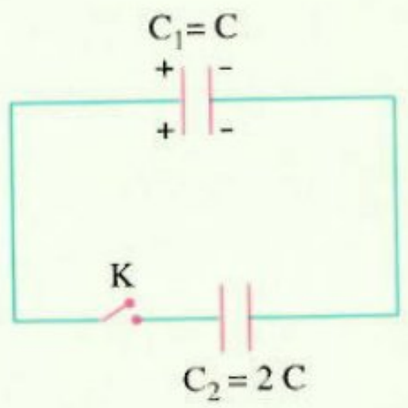
52 From the opposite electric circuit, the ratio between the total capacitance of capacitors before closing the key (K) and after closing it equals ..... . *(Exp. 23)*

(a) $\frac{7}{11}$ (b) $\frac{11}{7}$

(c) $\frac{6}{1}$ (d) $\frac{1}{6}$



53 The figure represents two capacitors (1) and (2), capacitor (1) is charged with a charge of 60 µC and capacitor (2) is uncharged. After closing the switch (K), which of the following choices represents the charge on the two capacitors (1) and (2)?



	The charge Q_1	The charge Q_2
(a)	40 µC	20 µC
(b)	20 µC	40 µC
(c)	30 µC	30 µC
(d)	Zero	60 µC

(1st session 23)

54 Three capacitors, each of capacitance (12 µF), were connected to a DC source of (20 V) with two different methods as in figures (A) and (B).

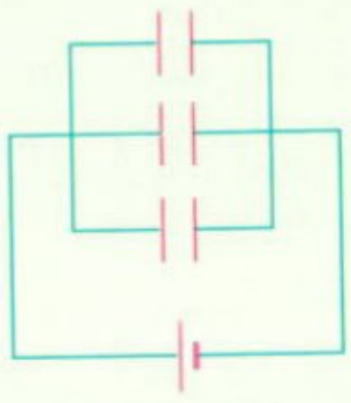


Figure (A)

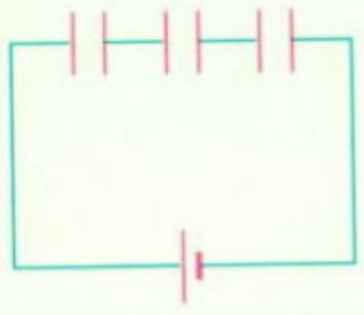


Figure (B)

CamScanner

Scanned with OKEN Scanner

CamScanner

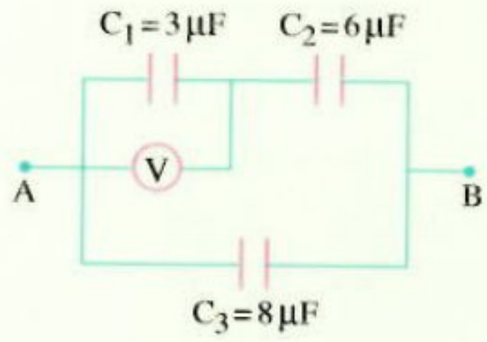
So, the ratio of the charge accumulated on each capacitor in circuit (A) to that in circuit (B) $\left(\frac{Q_A}{Q_B}\right)$ equals ..... .

(2nd session 23)

(a) $\frac{9}{1}$ (b) $\frac{1}{9}$ (c) $\frac{3}{1}$ (d) $\frac{1}{3}$

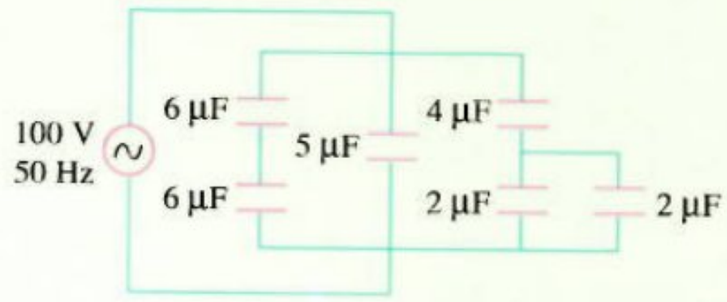
55 The opposite figure represents a part of an electric circuit. If the charge accumulated on one plate of capacitor (C_3) is 2.4 mC, then the voltmeter (V) reads ..... .

(Practice 25)



(a) 10 V (b) 20 V
(c) 100 V (d) 200 V

56 The figure shows an AC circuit consisting of an AC source and a set of capacitors. Based on the data recorded in the figure, the equivalent capacitive reactance in the circuit equals ..... .



(1st session 25)

(a) $\frac{4.8}{\pi}$ kΩ (b) $\frac{1}{\pi}$ kΩ (c) 2 π kΩ (d) π kΩ

57 An electric circuit consists of an alternating source of frequency $\frac{100}{\pi}$ Hz and a set of identical capacitors, each of capacitance C. When the capacitors were connected in series, the capacitive reactance in the circuit was 200 Ω. When the capacitors were connected in parallel, the capacitive reactance in the circuit was 12.5 Ω. So, the number of capacitors is ..... .

(2nd session 25)

(a) 4 capacitors (b) 8 capacitors (c) 12 capacitors (d) 16 capacitors

Scanned with OKEN Scanner